

REASONING & COMPUTER APTITUDE

Directions (1-3): Study the following information carefully and answer the given questions.

Ten persons Anu, Banu, Cathy, Dani, Emi, Fath, Guna, Hari, Imu and Jeni are born in five different months viz., January, August, April, June and December in the same year. No two persons whose names start with consecutive alphabet are neighbours. Two persons were born in one month such that ten persons were born in five months. They all are born on different dates viz.; each month consists of one odd-numbered date and one even-numbered date. The odd-numbered date ranges between 1 - 15 and the even-numbered date ranges between 16 - 30. Each one likes different fruits namely Papaya, Apple, Kiwi, Mango, Strawberry, Grapes, Banana, Guava, Pear and Plums, but not necessarily in the same order.

Anu was born in one of the months, which does not have 30 days. There were only two persons born between Anu and the one who likes Grapes. The one who likes Kiwi born immediately after Fath, who is not born in the first or last months of the year. There are three persons were born between Emi and the one who likes Banana, who is older than Emi. Fath and the one who likes Kiwi are born in the same month. Anu was born on the smallest odd-numbered date. The one who likes Guava were born on the 18th of August. As many persons born before the one who likes Mango is one more than the number of persons born after Emi, who does not like grapes. Fath and Anu do not like Mango. There are four persons were born between Imu and Cathy, who were born in April. Jeni was born on a date which is divisible by both 7 and 4. The one who likes Strawberry is not born in the same month as the one who likes Plums. Fath was born 200 days before Guna and Dani was born 3 days after Fath. The number of persons born between Hari and Dani is same as the number of persons born between Dani and Guna. Hari does not like Banana and is elder than the one who likes Mango. Guna does not born in the month which has 30 days. The one who likes Pear does not born in odd numbered date. More than three persons were born between Cathy and the one who likes Pear. The one who likes Papaya does not born in January and does not born in the same month as Guna was born. Guna and Hari are born in same dates. The one who likes Strawberry was born before Jeni and is not the eldest person. Hari does not born immediately after Anu. Cathy was born in smallest two-digit prime number. The one who like pear was born in largest possible even number. Emi and Banu were born on possible two-digit odd prime number.

- If Kiwi and Plum interchange their position, then who was born immediately after the one who likes Plum?
(a) Guna (b) Anu (c) Imu
(d) Cathy (e) None of these
- Which of the following is true about Hari?
I. He was born in the month having even numbered days.
II. Four people were born between Hari and Anu.
III. He likes Kiwi.
(a) Only I (b) Only II
(c) Only III (d) Both I and II
(e) Both II and III
- Who likes Grapes?
(a) The one who was born on April 28
(b) The one who was born immediately after the one who likes Banana

- The one who born before Fath
- Only I and III
- All I, II and III

Directions (4-6): Study the following information carefully and answer the given questions.

Two buses i.e., bus *P* and bus *Q* start their journey from a common depot. Both buses complete their individual journey. Bus *P* moved 7 km towards the south and reached stoppage 1. Then turned right and moved 8 km and reached stoppage 2. From there Bus *P* turned right and covered 9 km to reach stoppage 3. Then it turned to its left and covered 5 km to reach stoppage 4 and finally, it took a left turn and went 4 km to reach stoppage 5.

Bus *Q* went 6 km to the east and reached stoppage 6. Then it turned to its right and moved 3 km to reach stoppage 7. It travelled 13 km towards the left and went to stoppage 8. From there bus *Q* turned left and covered 8 km to reach stoppage 9. Finally, it moved towards its left and went 21 km to reach stoppage 10.

Now these stoppages are assigned names according to the below given conditions:

'W' is assigned to the stoppage if the distance between the consecutive stoppages is a prime number.

'X' is assigned to the stoppage which is in the north-west or south-west direction of the depot.

'Y' is assigned to the stoppage which is in the north-east or south-east direction of the depot.

'Z' is assigned to the stoppage if the distance between the stoppage and the depot is an even number.

Note: Multiple names can be assigned to a single stoppage.

- Bus *X* starts the journey from the depot and goes 5 meters south and takes a left turn and travel 10 meters. Then it takes a left turn and reaches somewhere between stoppage 9 and 10. What will be the distance from that point to stoppage 9?
(a) 16 meters (b) 9 meters
(c) 10 meters (d) 13 meters
(e) 15 meters
- Four of the following five are alike in a certain way and so form a group. Which one does not belong to that group?
(a) Stoppage 5 - Stoppage 10 (b) Stoppage 3 - Stoppage 9
(c) Stoppage 2 - Stoppage 6 (d) Stoppage 1 - Stoppage 10
(e) Stoppage 5 - Stoppage 6
- Which of the following stoppage assigned maximum names?
(a) Stoppage 3 (b) Stoppage 8
(c) Stoppage 9 (d) Stoppage 7
(e) Stoppage 1

Directions (7-10): Study the following information carefully and answer the given question.

21 persons work in a bank in a different post. Posts are in decreasing order i.e., ED, DGM, AGM, Manager, AM and Clerk. It is known that on any post the number of persons posted is one more than the number of persons posted in the immediate senior post. For example: If 4 person works as AM then 5 person works as Clerk.

E, H and D work at the same post. S is senior to V and C. Neither X nor Q is a manager. P, L, M and W work at the same post. V is just senior to X. C is senior to A. U and T are senior to L. R is senior to T but junior to F and G. V is just junior to N. R and N do not work at the same post. Q is senior to W. B is senior to O. C and Q works at the same post. A does not work as clerk, P does not work as AM and T does not work as AGM. S is senior to R but junior to G. Now these persons are transferred to three departments i.e., HR, Marketing and Legal using the below-given conditions.

I. No transfer happens for the person who belongs to the two highest positions and they become leaders in legal, Marketing and HR respectively according to the alphabetical order.

II. Persons whose name starts with a letter that comes before M in alphabetical order and he/she works with a person whose name starts with a vowel went to HR (excluding the vowels).

III. Persons whose name starts with a letter which comes after M in alphabetical order and he/she works at a post which has even number of persons went to Marketing.

IV. Rest will go to the Legal department.

7. Four of the following five are alike in a certain way and so form a group. Which one does not belong to that group?

- (a) A (b) D (c) E
(d) F (e) H

8. Which of the following is true?

- (a) At least two of the clerks is transferred to the legal department.
(b) At least one of the AMs is transferred to marketing department.
(c) At least one of the AGMs is transferred to HR department.
(d) At least one of the DGMs is transferred to marketing department.
(e) At least one of the Managers is transferred to HR department.

9. Who among the following is not the manager?

- (a) B (b) C (c) N
(d) T (e) U

10. In which department is U transferred to?

- (a) HR (b) Marketing
(c) Legal (d) Either HR or Legal
(e) None of these

Directions (11-13): Study the following information carefully and answer the given questions.

In a certain code of language,

‘one apple daily please’ is written as ‘A\$ A@ B% B@’.

‘doctor advised help better’ is written as ‘B% B& A^ B%’.

‘almost all treat well’ is written as ‘A% B@ A\$ B&’.

‘pollute air thus rivers’ is written as ‘B^ A& B& B%’.

11. What is the code of “SCHOOL”?

- (a) B\$ (b) A& (c) A\$
(d) B% (e) None of these

12. What is the code of ‘America Africa’?

- (a) A^ B% (b) A@ A* (c) A^ A%
(d) B% B^ (e) None of these

13. Which of the following is the code for ‘Rain Starts Even’?

- (a) B& B% A& (b) A* A@ A&
(c) B& B% A@ (d) B% B* A\$
(e) None of these

14. **Statement:** A cotton dress manufacturing company Z has planned to import the materials for its production from Country A which is a leading cotton producer in the world and the top exporter of Cotton materials. The manufacturing company Z has ordered material in huge quantities worth \$45 million because of the increasing demand for cotton-based dress material in its company.

Which among the following option(s) will weaken the given statement?

- I. Company Z has earned more profit this year when compared to the previous two years.
II. People do not like to wear cotton dresses in the winter season.
III. Country A took most of the orders for the cotton materials when compared to Country B. (IBPS PO Mains 2022)
(a) Only I (b) Only II and III
(c) Only III (d) Only I and III
(e) Only II

Directions (15-16): Study the information given carefully and answer the question given below.

Ten people A, B, C, D, E, L, M, N, O, and P are sitting around a circular table but not necessarily in the same order. Five of them are facing inside while others are facing outside. No two persons, whose names start with consecutive alphabets sit adjacent to each other.

E sits immediate right of the one who faces D. B does not face L. One person sits between C and M. Immediate neighbours of L are facing outside. Two persons sit between B and D. O faces the same direction as B. Both D and N face opposite directions as P. E sits fourth to the right of L. A is an immediate neighbour of C, who faces outside. One person sits between E and A. Both immediate neighbours of A face opposite directions to each other. Neither P nor O sits adjacent to B. E sits third to the left of P. A does not sit second to the right of E.

15. What is the position of P with respect to C?

- (a) Fifth to the right (b) Fourth to the right
(c) Second to the left (d) Second to the right
(e) Third to the left

16. Who sits opposite to D?

- (a) C (b) M (c) A
(d) E (e) L

Directions (17-20): Study the following information carefully and answer the given questions.

Nine people A, B, C, D, E, F, G, H, and J live in a nine-storey building such that the bottommost floor is numbered as 1 and the topmost floor is numbered as 9 but not necessarily in the same order. They can speak nine different languages – Assamese, Bengali, Gujarati, Odia, Sanskrit, English, Telegu, Hindi, and Kannada but not necessarily in the same order.

H cannot speak Kannada. F lives above H. Two persons live below G. The number of persons, who live above the person, who speaks Sanskrit is equal to the number of persons, who live below A. G speaks either Telegu or Hindi. E's floor number is half of C's floor number. One person lives between G and the one, who speaks Sanskrit. D lives immediately above J. D neither speaks Telegu nor Kannada. One person lives between A and the one, who can speak Gujarati. The one, who can speak Assamese does not live on a prime numbered floor. C does not live immediately above or immediately below G's floor. The one, who speaks English lives three floors above J. One person lives between C and B, who can speak Odia. The one, who can speak Assamese lives immediately above the, one who can speak Bengali.

17. Who lives immediately above the one, who can speak Sanskrit?

- (a) H (b) E (c) F
(d) A (e) B

18. Which of the following combination is correct?

- (a) 3rd floor - G - Hindi (b) 1st floor - D - Telegu
(c) 5th floor - H - Kannada (d) 7th floor - F - Gujarati
(e) 9th floor - F - Assamese

19. How many persons live between A and C?

- (a) Five (b) Four (c) Three
(d) Two (e) One

20. Who among the following persons speaks Gujarati?

- (a) H (b) F (c) D
(d) J (e) E

Directions (21-23): Study the following information carefully and answer the questions.

In each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and Give answer:

21. What is the code for "Coming"?
- In a certain code language, "Holidays are fun" is coded as "pnj kol sip", "winter holidays coming" is coded as "kol bit pop" and "Winter is cold" is coded as "bit fog sap".
 - In a certain code language, "They are coming" is coded as "pop top gig" and "How are they" is coded as "top sun gig".
- Only I is sufficient
 - Only II is sufficient
 - Both are not sufficient to answer the question
 - Both are necessary to answer the question
 - Either I or II is sufficient
22. Six people A, B, C, D, E, and F are sitting in a row facing north.
- A sits second from one end. One person sits between A and B. C sits immediately left of D. F sits to the right of C.
 - B sits third from one of the ends. F who does not sit at the corner is one of the positions right to B. One person sits between D and F. Two people sit between D and A.
- Only I is sufficient
 - Only II is sufficient
 - Both are not sufficient to answer the question
 - Both are necessary to answer the question
 - Either I or II is sufficient
23. X is in which direction with respect to Q?
- P is 5 meters west of Q. R is 10 meters south of Q. S is 8 meters east of R. T is 12 meters north of S. X is 8 meters west of T.
 - V is 8 meters south of Q. V is 4 meters east of U which is 10 meters south of W. Y is 2 meters west of W. Distance between X and Y is 2 meters.
- Statement I is alone sufficient
 - Statement II is alone sufficient
 - Both statements are together necessary to answer the question
 - Both statements are together not sufficient to answer the question
 - Either I or II is sufficient to answer the question.

Directions (24-26): Study the information given carefully and answer the questions given below.

$P @ Q$ means P is the father of Q.

$P \# Q$ means P is the mother of Q.

$P \$ Q$ means P is the husband of Q.

$P \% Q$ means P is the wife of Q.

$P \& Q$ means P is the son of Q.

$P * Q$ means P is the daughter of Q.

$P = Q$ means P is the sister of Q.

$P + Q$ means P is the brother of Q.

24. In the expression 'R # P = S # Q + J (?) T; R % M @ S % T', what will come in the place of the '?' to satisfy J is the child of T?
- &
 - *
 - +
 - Either (a) and (b)
 - Either (b) or (c)

25. In the expression 'M * S \$ R # A + B \$ N'. How N is related to M?
- Sister
 - Brother
 - Cousin
 - Daughter
 - Sister-in-law

26. In the expression 'P % B @ N = Q \$ S # T; Q + M @ C'; how is T related to C?
- Sister
 - Brother
 - Cousin
 - Sister-in-law
 - Cannot be determined

27. In the question below are given a statement followed by two courses of action numbered I, II and III. A course of action is a step or administrative decision to be taken for improvement, follow-up or further action in regard to the problem, policy etc. On the basis of the information given in the statement to be true, then decide which of the suggested courses of action logically follow(s) for pursuing.

Nowadays children are more attracted towards playing mobile and computer games rather than playing outdoor games. These games offer a remarkable graphical interface with illuminating images of cars, soldiers, fighters, and players which attracts the interest of children towards this technology and there are thousands of games on one platform. But it has a very bad effect on children's health. It leads to obesity, muscular strain, and eye damage. Overall, this is not a beneficial trend as it can isolate a child from society and make them unhealthy.

Course of actions:

- Children should play outdoor games to keep their bodies and mind fit and fine.
 - Parents should stop using mobile so that the children could not play mobile games.
 - Parents should tell their children to focus only on study not any kind of games.
- Only II and III follow
 - Only I follows
 - Only II follows
 - Only I and III follow
 - All I, II and III follow

Directions (28-29): Study the following information and answer the following questions.

Cognizant has reduced the maximum bench time for employees, opening up the possibility of higher numbers of exits in the coming months. The company has shrunk the bench time from 60 days to 35 days for those employees who aren't on billable projects. After 35 days, they could be asked to leave. The formal exit process may take another 60 to 90 days.

- Cognizant pays its employees even when they are not on billable projects.
 - More layoffs will take place in Cognizant in the coming days.
 - Headcounts in Cognizant are definitely going to shrink in the days to come globally.
 - The company is trying to reduce its overhead expenses incurred on the employees.
 - Pay and perk is set to decrease for Cognizant employees in the coming years.
28. Which of the following among (A), (B), (C), (D) and (E) has been assumed in the given information?
- Only B
 - Only C
 - Only D
 - Only A
 - Only E
29. Which of the following among (A), (B), (C), (D) and (E) will be the effect of the reduction in the bench time by Cognizant?
- Only B
 - Only D
 - Only A
 - Only C
 - Only E

Directions (30-33): Study the following information carefully and answer the given questions.

9 persons P to X belong to three different states (Tripura, Manipur, and Assam) of different ages. Not more than 4 and not less than 2 persons belong to each state. If it is given for example B and the one who is 10 years belong to the same state then that means B is not 10 years.

P and V do not belong to the same state. S does not belong to Tripura and Assam. Q and the one who is 18 years old belong to the same state and these are the only persons belonging to that state. U and the one who is 29 years old belong to the same state and U 's age is an odd number. W and the one who is 54 years old belong to the same state. All V , the person one who is 61 years old and the person one who is 64 years old belong to the same state. P is the eldest person in his/her respective state. The one who is 53 years belongs to Tripura. S age is twice of Q . V 's age is 7 years less than W 's age. R is not the youngest person. The minimum age is 18 years. V is elder than Q . P and S do not belong to the same state. V and T belong to the same state. U is 5 years elder than Q and one of the ages is 51 years.

30. What is the age of V ?

- (a) 37 (b) 54 (c) 35
(d) 29 (e) 61

31. Who among the following belongs to Assam?

- (a) P (b) R (c) T
(d) U (e) X

32. The sum of the ages of R and S is:

- (a) 125 (b) 59 (c) 88
(d) 121 (e) 53

33. Four of the following five are alike in a certain way and so form a group. Which one does not belong to that group?

- (a) S (b) T (c) U
(d) V (e) W

Directions (34-36): In the following questions, the symbols $\#$, $\$$, $+$, $@$ and $*$ are used with the following meaning as illustrated below:

' $A \# B$ ' means ' A is either smaller than or equal to B .'

' $A \$ B$ ' means ' A is neither smaller than nor equal to B .'

' $A + B$ ' means ' A is either greater than or equal to B .'

' $A @ B$ ' means ' A is neither greater than nor equal to B .'

' $A * B$ ' means ' A is neither greater than nor smaller than B .'

In the following question, assuming the given statements to be true, find out which one of the two conclusions I and II given below is/are definitely true.

34. **Statements:** $S @ T \$ Q + R$; $Q * P @ W$

Conclusions:

- I. $R \# W$
II. $W \$ R$
(a) Only I follows (b) Only II follows
(c) Both I and II follow (d) Either I or II follows
(e) Neither I nor II follows

35. **Statements:** $S @ T \$ Q + R + U$; $Q * P \# W \# V$

Conclusions:

- I. $V \$ U$
II. $U \# V$
III. $R @ T$
(a) Only II follows (b) Only I and II follows
(c) Only II follows (d) Only II and III follows
(e) Only I follows

36. **Statements:** $U @ V + W \# Y * X$; $V + P \$ Q$

Conclusions:

- I. $P \# X$
II. $Q @ U$
III. $W \# X$

- (a) Only II follows (b) Only III follows
(c) Only I follows (d) All follows
(e) None follow

37. In the question the statement is given followed by some inferences. You must consider the statement to be true even if it seems to be at variance from commonly known facts.

There is the considerable reduction in the number of people affected by water-borne diseases in City X during this rainy season. Waterborne diseases are caused by drinking contaminated or dirty water. Contaminated water can cause many types of diarrheal diseases, including Cholera, and other serious illnesses such as Guinea worm disease, Typhoid, and Dysentery. The government has opened four new civil hospitals in City X in the beginning of the year to treat these diseases by providing better Health Services. These hospitals strive to help people improve their productivity and reduce risks of diseases and injury in a cost-effective way.

Which of the following statement can be inferred from the passage?

- (a) The number of people affected by waterborne diseases has reduced over the past few years.
(b) The hospital helps poor people who were suffering from waterborne diseases with better health services.
(c) The Government opened new hospitals to provide cost-effective health services.
(d) As people spend more money to treat this disease, the Government opens new hospitals to provide treatment in a cost-effective way.
(e) Drinking contaminated water causes waterborne diseases including diarrheal disease.

38. **Statement:** Prime Minister Narendra Modi is all set tomorrow to launch three mega flagship schemes aimed at transforming urban India, including the much talked about Smart Cities proposal, with a whopping Rs 4 lakh crore kitty having been lined up for the projects. Urban Development Minister States and local bodies will play a key role in the execution of these schemes, the minister said, adding that there will also be a lot of scope for private participation.

"Effective management of growing urbanization is a major challenge towards the making of resurgent India. Resource management is a major challenge. The government, through these schemes, is looking to bring about a transformation of urban India with the critical goal of making the country an economic hub. Speaking at the event, Modi said, "The decision to make cities smart will be taken not by governments but by the people of the city, the local administration. Let there be competition between the cities in that smart cities come up".

Which of the following will be an effect of the initiative taken by the government?

- (a) Millions of jobs will be created and scarcity of houses will be removed in urban and rural areas.
(b) Quality of life will improve in rural and urban areas.
(c) Enhanced citizen and government engagement besides the implementation of reforms in urban governance would be the focus under the scheme.
(d) Ensuring water supply to each household, sewer connections, sanitation and solid waste management in rural and urban governance would be the focus under the scheme.
(e) All of the above

39. **Statement:** Beijing is locking down to contain the pandemic and cracking down on economically critical sectors and on high corporate debt with an aggression unmatched by any other government. That goes a long way to explain why China is slowing so fast now when the rest of the world is not. But the link between growth in China and other economies started to loosen about five years ago, so this moment may reflect deeper forces in play.

Which of the following can be inferred from the passage?

- I. Most of the economies in the world were growing in close sync with China about five years ago.

- II. The Pandemic has created the decoupling factor between China and the rest of the world.
- III. China has changed its policy stance pertaining to the economy which has led its GDP to fall.
- (a) Both II and III (b) Only I
(c) Both I and III (d) Only III
(e) All of the above

Directions (40-44): Study the following information carefully and answer the given questions.

Ten persons sit in two parallel rows with 5 persons in each row. Row 1 has A, B, C, D and E facing north and row 2 has P, Q, R, S and T facing south. These persons belong to a three-generation family.

A sits 2nd from one of the ends. R faces the immediate neighbour of A's niece. E's mother-in-law sits opposite to the one who sits 3rd to the right of A. One person sits between A and her sister-in-law. B is married to E who has one child. R sits to the right of T. Q is married. A's son sits immediate right of A. B's father faces A who has two children one son and one daughter. P and her granddaughter Q are not sitting together. Three persons sit between B and the person sitting opposite to Q's father. C and D are unmarried. C does not sit next to E.

40. How C is related to Q?
(a) Mother (b) Son (c) Cousin
(d) Sister (e) None of these
41. Four of the following five are alike in a certain way and so form a group. Which one does not belong to that group?
(a) S (b) R (c) P
(d) C (e) B
42. How T is related to A?
(a) Son (b) Husband (c) Brother
(d) Father (e) Son-in-law
43. Who sits opposite to the one who sits immediate left of R?
(a) A (b) B (c) C
(d) D (e) E
44. How P is related to E?
(a) Father-in-law (b) Mother-in-law
(c) Sister-in-law (d) Daughter
(e) Daughter-in-law
45. Which of the following word(s) end with a vowel, if all the letters in the three words given below are changed to the previous third letter as per the English alphabetical order and then are arranged in the alphabetical order from left to right within each word?
- I. CONFIDENT
II. VOCAB
III. EXERCISE
- (a) Only I (b) Only II
(c) II and III only (d) III only
(e) None

ENGLISH LANGUAGE

Direction (46-49): In each question a sentence is given followed by a blank. You are provided with five options. You have to identify which statement/statements can fill in the blank and carry forward the given sentence in the most logical way so as to make the sentence coherent and contextually correct.

46. We urge the government to have a serious go at such improvement, and the result would, in tandem with the _____ that relieves banks of the risk of funding long-gestation projects they are ill-suited to finance, create well-capitalized banks that are answerable to savvy shareholders on the board.

- (a) choice of capital infusion stocking and trading
(b) creation of a vibrant market for corporate bonds
(c) opting for public issues to shore market
(d) optional funding
(e) availability of the capital

47. The writing off of a bad loan means that the bank would have to make 100% provisioning (setting aside capital) _____ even if that loan is not repaid, and clean up its books.

- (a) considering the payoff
(b) maintaining the contingencies
(c) against the net worth of the company
(d) against the loan to protect depositors
(e) assuring the payoff of the bad loans

48. After all, broad-based shareholding is feasible as banking is seen as a safe business: banks have a guaranteed _____, and face limited competition.

- (a) loan payoff strategy
(b) partnership strategic investment
(c) control over the infrastructural database
(d) Growth-investing strategy
(e) clientele besides a growing depositor base

49. Raising equity from the public will, besides giving the government fiscal room, help lower State ownership below 51%, giving PSBs leeway to step outside the shadow of vigilance supervision and _____.

- (a) write-offs for banks are inevitable
(b) bankruptcy code is put to use for the resolution of bad loans
(c) to fix their own remuneration plans
(d) RBI has warned that the modest NPA
(e) veils the strong undercurrent of slippage

Directions (50-52): In the following question, a part of the sentence has been highlighted. You need to identify which of the given options can replace the highlighted part and make the sentence grammatically and contextually correct. In case the highlighted part is correct, mark (e) as your answer.

50. Last month's regulatory green signal for Indian Gas Exchange (IGX), the first national-level gas trading platform, **is welcome reforming that would shore up transparent price** discovery in natural gas, the cleanest and most efficient fossil fuel.

- (a) is welcoming reform that would shore transparent price
(b) is welcome reform that would shore up transparent price
(c) is welcome reform that would shore transparent price
(d) is welcome reform that would shoring up transparent price
(e) None of these

51. The case to make reverse charge universal for all input purchases by firms — so that buyers **pay tax on their purchases directly into the government** while recognizing the vendor on whose behalf the tax is being paid — is compelling.

- (a) pay tax on their purchases directly the government
(b) pay taxation on their purchases directly to the government
(c) pay tax on their purchases directly to the government
(d) pay tax on their purchases directly belonging the government
(e) None of these

52. As evident in the ongoing and overlong quarrel between farmers agitating against the new farm laws and GoI, providing **enlarge incentives can be perceived as alarming disincentives**.

- (a) enriching incentives perceived as alarming disincentives
(b) enriching incentives can perceived alarm disincentives
(c) enlarge incentive can bring alarmed disincentives
(d) enriching incentives can be perceived as alarming disincentives
(e) None of these

Directions (53-55): In the following questions, two words are highlighted in a sentence. Choose the most appropriate pair of SYNONYMS/ANTONYMS of the highlighted words. If none of the alternatives is correct, choose option E as your answer.

53. A New York judge has **upheld** an order preventing The New York Times from publishing documents between conservative group Project Veritas and its lawyer and ruled that the newspaper must immediately **relinquish** confidential legal memos it obtained.
 (a) praise: justify (b) supported: abdicate
 (c) advocate: represent (d) promote: foster
 (e) None of these
54. U.S. President Joe Biden signed into law legislation that bans imports from China's Xinjiang region over concerns about forced labor, the White House said, **provoking** an angry Chinese **condemnation**.
 (a) baffle: uncertain (b) abandon: divulge
 (c) celebrate: discriminate (d) appease: acclaim
 (e) None of these
55. Japanese Prime Minister Fumio Kishida's government on Friday approved record defence spending, with a 10th straight annual increase in 2022, against a backdrop of China's **rapid** military **expansion** and North Korea's nuclear and missile programmes.
 (a) widespread: available (b) ameliorate: effective
 (c) unhurried: reduction (d) distinct: increase
 (e) None of these

Directions (56-60): In the following passage, some of the words are given in bold, each of which is preceded by a blank. Find a suitable word from the options given below that could replace the bold word so as to make the paragraph meaningful. If no replacement is required, choose (E), as your answer.

Journalism's obituary has been written quite a few times. For instance, in the modern world, the _____ (**snowballing**) (A) revenues of traditional journalistic platforms, the print media, for instance, intimidatory intervention, legal or otherwise, against digital news networks to stifle free speech and, equally important, the _____ (**receptiveness**) (B) of the media fraternity before the powers that be — be it the government or big business are being cited as the principal causes that would lead to journalism's demise. Yet the _____ (**ensconce**) (C) has turned out to be a rather stubborn species, proving the pundits wrong repeatedly. The conferring of this year's Nobel Peace Prize on two courageous, proficient scribes, Maria Ressa of the Philippines and Russia's Dmitry Muratov, shows that journalism can, indeed, be _____ (**reverend**) (D) in spite of such formidable threats as State repression as well as the public vulnerability to enchantment with false narratives. The path to renewal for individual journalists and news organizations, as Ms Ressa and Mr Muratov have shown, lies in the ability to refresh journalism's commitment to the original charter of the fourth estate: to report the truth without fear or favour and uphold the primacy of free speech. This is, of course, not a road for the faint-hearted. Real journalistic work is considered to be a dangerous profession: in 2020, a report compiled by the director-general of UNESCO found that only 13 percent of cases filed against newsmen around the world had been resolved the figures were 12 and 11 percent, respectively, for the previous two years. _____ (**emancipation**) (E), intimidation, threats of bodily violence and trolling these are especially true for women journalists and even death. India is witness to Gauri Lankesh's murder, is the 'occupational hazards' in the journalistic tryst with truth.

56. Which of the following options will come in blank A?
 (a) Mortifying (b) Aggrandizing
 (c) Plummeting (d) Demeaning
 (e) No replacement required

57. Which of the following options will come in blank B?
 (a) Catcalling (b) Capitulation
 (c) Calumny (d) Resistance
 (e) No replacement required
58. Which of the following options will come in blank C?
 (a) Vocation (b) Usurpation
 (c) Avocation (d) Incumbency
 (e) No replacement required
59. Which of the following options will come in blank D?
 (a) Reorchestrated (b) Revalorized
 (c) Reconsecrated (d) Resurrected
 (e) No replacement required
60. Which of the following options will come in blank E?
 (a) Manumission (b) Transliteration
 (c) Incarceration (d) Annihilation
 (e) No replacement required

Directions (61-63): In the following questions few sentences are provided. Identify and mark the sentence which may fail to become part of the paragraph coherently. In questions where "none of these" is an option and all the sentences are meaningful and logical with the context of the paragraph mark (e) i.e., "none of these" as your answer choice.

61. (a) There are so many things going on in the world, our daily personal lives, faced-paced working environments and social media, that it can become overwhelming. It is necessary to stop amid all this and just ask yourself, "How am I doing?". If you are stressed, distracted, sad or cannot focus on work, maybe it is time you take a break, drink a big glass of water, take a walk or breathe for a moment.
 (b) Explaining why we need self-care, Dr Samridhi said, "Mental health includes all aspects of one's life - physical, emotional or psychological wellness and social well-being. And self-care can play a vital role in enhancing our well-being and maintaining our mental health."
 (c) Failing to address the mental health condition of an adolescent leads to numerous consequences that even extend to adulthood. It impairs both physical and psychological health. Many parents find it challenging to tackle the mental health-related disorder of their children, and some even fail to identify that their child suffers from such a disorder. The mental illness that a child face in adolescence is utterly different from adulthood mental challenges.
 (d) Taking offs when you feel unmotivated is a luxury not many of us have, but it definitely isn't poor ethics. It is a basic human need that all of us have and need to function adequately. And that is where self-care comes in. We must take a few hours from our daily schedule and focus on taking time off to care for our mental health.
 (e) None of these
62. (a) Eating disorder among teenagers is a prevalent practice. It involves abnormal eating practices and preoccupation with food, which leads the adolescent to be concerned about their body shape and weight. Suddenly leaving an amount of food consumption can impact the physical and mental health severely. Such disorders can extend to adulthood if not taken care of in the early stages.
 (b) Dysbiosis (increase in unhealthy bacteria in the gut) affects infant and toddler behaviour such as feeling scared or fearful, feeling depressed or disconnected, irritability and anxiety/nervousness and cognitive development such as learning, speech, concentration and analytical thinking.
 (c) We use phrases such as "follow your gut" or "listen to your gut" quite often when we have to make crucial decisions. Have you felt "butterflies in your stomach" or "a numbing feeling in your stomach" when you are anxious? Well, you are likely

getting signals from an unexpected source: your second brain — the gut. Gut health can directly influence the expression and management of our mental health including depression, anxiety and stress.

- (d) The gut microbiome is the community of microbes and their genes that reside within the gastrointestinal tract and are a key driver of neurobiological and behavioural development. Research on the human microbiome clearly links dynamics in the gut with neurobiological development in children.
- (e) None of these

63. (a) “NASA takes the cost and schedule commitments of its projects and programs very seriously. We are exploring options for the mission in the context of the Discovery Program, and a decision on the path forward will be made in the coming months” said Thomas Zurbuchen, associate administrator for NASA’s Science Mission Directorate, in a press statement.
- (b) Data science tools are one of the hottest tools to use for effective data management to derive meaningful insights for better customer engagement. Multiple data and tech companies including FAANG (Facebook, Apple, Amazon, Netflix, Google) are popular for letting data scientists leverage different data science tools. Tech companies and FAANG need to have a deep understanding of customer behavior in this data-driven market. Thus, aspiring data scientists should look out for the top data science tools to get recruited by popular tech companies including FAANG in 2022 and beyond.
- (c) Psyche was selected in 2017 as part of NASA’s Discovery Program, which is a line of low-cost competition missions led by a single principal investigator. The agency is now forming an independent assessment team to review the path forward for the project and the Discover Program as a whole.
- (d) NASA has announced that the Psyche mission, the space agency’s first mission designed to study a metal-rich asteroid, will not be attempting a 2022 launch. The late deliver of the spacecraft’s flight software and testing equipment means that NASA does not have enough time to complete the testing needed ahead of its remaining launch period this year which ends on October 11. The mission team needs more time than that to ensure that the software will function properly in flight.
- (e) None of these

Directions (64-65): In the following questions, five statements are given out of which one is wrong. Choose the one that consists of an error and mark that as your answer.

64. A. The UN, which has prevaricated on Syria, has estimated that more than 80 percent of Syrians live below the poverty line.
- B. Bashar al-Assad, the beleaguered President of Syria, lives to fight another day, having been sworn in as President for a fourth term on Saturday.
- C. One major reason for the Soviet Union suffering an economic bust in 1991 was attributed against the Kremlin.
- D. President Assad reaffirms his grip on a direly depleted inheritance with a welter of problems that could impede his foreign and domestic policies.
- E. All are true
- (a) D (b) C (c) B
- (d) A (e) E
65. A. Another political problem that has contributed to the melting of Arctic ice sheets are the improved navigability of the northern oceans.
- B. The world watches as the United States withdraws from Afghanistan leaving the country in the throes of civil war.
- C. The states of North Rhine-Westphalia, Rhineland-Palatinate and Saarland have been the worst affected by the record rainfall, which authorities have called the heaviest in a century.

D. President Xi Jinping’s government in Beijing to be “more transparent” even as the medical fraternity has been groping in search for the origins.

E. Across the Atlantic, the weather forecasts have been less than convincing and that has added to the woes of people.

- (a) A (b) B (c) C
- (d) D (e) E

Directions (66-73): The passage given below is followed by a set of questions. Choose the best answer to each question.

The human brain is an amazing computing machine. Weighing only three pounds or so, it can process information a thousand times faster than the fastest supercomputer, store a thousand times more information than a powerful laptop, and do it all using no more energy than a 20-watt light bulb. Researchers are trying to replicate this success using soft, flexible organic materials that can operate like biological neurons and someday might even be able to interconnect with them. Eventually, soft “neuromorphic” computer chips could be implanted directly into the brain, allowing people to control an artificial arm or a computer monitor simply by thinking about it. Like real neurons but unlike conventional computer chips these new devices can send and receive both chemical and electrical signals. “Your brain works with chemicals, with neurotransmitters like dopamine and serotonin. Our materials are able to interact electrochemically with them,” says Alberto Salleo, a materials scientist at Stanford University who wrote about the potential for organic neuromorphic devices in the 2021 Annual Review of Materials Research. The work grows out of an increasing interest in neuromorphic computer circuits that mimic how human neural connections, or synapses, work. These circuits, whether made of silicon, metal or organic materials, work less like those in digital computers and more like the networks of neurons in the human brain.

Conventional digital computers work one step at a time, and their architecture creates a fundamental division between calculation and memory. This division means that ones and zeroes must be shuttled back and forth between locations on the computer processor, creating a bottleneck for speed and energy use. The brain does things differently. An individual neuron receives signals from many other neurons, and all these signals together add up to affect the electrical state of the receiving neuron. In effect, each neuron serves as both a calculating device _____ the value of all the signals it has received and a memory device: storing the value of all of those combined signals as an infinitely variable analog value, rather than the zero-or-one of digital computers. Researchers have developed a number of different **memristive** devices that mimic this ability. When you run electric currents through them, you change the electrical resistance. Like biological neurons, these devices calculate by adding up the values of all the currents they have been exposed to. And they remember through the resulting value their resistance takes.

A simple organic memristor, for example, might have two layers of electrically conducting materials. When a voltage is applied, electric current drives positively charged ions from one layer into the other, changing how easily the second layer will conduct electricity the next time it is exposed to an electric current. “It’s a way of letting physics do the computing,” says Matthew Marinella, a computer engineer at Arizona State University in Tempe who researches neuromorphic computing. The technique also liberates the computer from strictly binary values. “When you have classical computer memory, it’s either a zero or a one. We make a memory that could be any value between zero and one. So you can tune it in an analog fashion,” Salleo says.

At the moment, most memristors and related devices aren’t based on organic materials but use standard silicon chip technology. Some are even used commercially as a way of speeding up artificial intelligence programs. But organic components have the potential to do the job faster while using less energy, Salleo says. For instance, Francesca Santoro, an electrical engineer now at RWTH Aachen University in Germany, is developing a polymer device that takes input from real cells and “learns” from it. In her device, the cells are separated from the artificial neuron by a small space, similar to the synapses that separate real neurons from one

another. As the **artificial (1)** produces **electrical (2)**, a nerve-signaling chemical, the dopamine changes the **dopamine (3)** state of the **cells (4)** half of the device. The more dopamine the cells produce, the more the electrical state of the artificial neuron changes, just as you might see with two biological neurons. "Our ultimate goal is really to design electronics which look like neurons and act like neurons," Santoro says.

66. "Your brain works with chemicals, with neurotransmitters like dopamine and serotonin. Our materials are able to interact electrochemically with them," says Alberto Salleo, a materials scientist at Stanford University who wrote about the potential for organic neuromorphic devices in the 2021 Annual Review of Materials Research. This statement information is expressed and communicated in the passage by which one of the scientists?
- (a) Francesca Santoro (b) Salleo
(c) Matthew Marinella (d) Alberto Salleo
(e) None of these
67. Which of the following is a suitable word to fill the blank as given in the passage?
- (a) Integrating (b) Implementing
(c) Segregating (d) Splitting
(e) Submitting
68. Which of the following is/are TRUE according to the passage?
- (a) The development is a result of the growing fascination with computer circuits called neural synapses. These circuits, whether composed of silicon, metal, or organic materials, use the human brain more than they do like those in digital computers.
(b) A fundamental distinction between calculation and memory is made by the architecture of traditional digital computers, which operate in steps. Because of this division, ones and zeroes must be moved back and forth between locations on the computer processor, slowing down processing and consuming more energy.
(c) The work is a result of growing research into computer circuits called neuromorphic synapses, which are designed to imitate the functioning of human neural connections. No matter if they are built of silicon, metal, or organic materials, these circuits function more like the networks of neurons in the human brain than like those in digital computers.
(d) Both (a) and (b)
(e) Both (b) and (c)
69. Which of the following is the appropriate title for the passage?
- (a) Channelising the process information
(b) Developing computer chips which act more like brain cells
(c) Brain working and neurotransmitters
(d) Developing electrochemical materials
(e) Implementing neuro morphic computer circuits
70. Which of the following is an appropriate word that can replace the word 'memristive' as given in bold in the passage?
- (a) Futuristic (b) Calculative
(c) Cognitive (d) Conductive
(e) None of these
71. According to the given passage, the brain does things differently and which of the statements is/are explain the functioning of the brain in effect with each neuron?
- (a) Numerous neurons send signals to a single neuron, and the sum of all these messages affects the electrical state of the receiving neuron.
(b) Different processes occur in the brain. Numerous neurons send messages to a single neuron, and when all of these signals combine, they have an impact on the receiving neuron's electrical state.

(c) Each neuron functions as a memory device that stores the value of all the combined signals it has received as an indefinitely variable analog value rather than the zero-or-one of digital computers as well as a device that calculates the value of all the signals it has received.

- (d) All of these
(e) All except (c)

72. Matthew Marinella, a computer engineer at Arizona State University in Tempe who researches neuromorphic computing has done research on various studies, which of the following statements is expressed and communicated in the passage by him?

- (a) When you have classical computer memory, it's either a zero or a one. We make a memory that could be any value between zero and one. So you can tune it in an analog fashion.
(b) Our ultimate goal is really to design electronics which look like neurons and act like neurons.
(c) When a voltage is applied, electric current drives positively charged ions from one layer into the other changing how easily the second layer will conduct electricity the next time it is exposed to an electric current. It's a way of letting physics do the computing.
(d) All of these
(e) None of these

73. In the given passage in one of the statements four words are given in bold. These four words may or may not be in their correct position. Find the correct combination of the words to make the sentence grammatically and contextually correct. If the sentence is correct as it is then select option (e) as your choice.

As the **artificial (1)** produces **electrical (2)**, a nerve-signaling chemical, the dopamine changes the **dopamine (3)** state of the **cells (4)** half of the device.

- (a) 1-4 and 2-3 (b) 2-4 and 1-3
(c) Only 1-2 (d) Only 1-3
(e) None of these

Directions (74-79): A passage has been given below. Some words and phrases have been highlighted to make them easily locatable while answering the questions. Read the following carefully and answer the questions that follow.

There was a time when the US used food aid as a powerful diplomacy tool to contain hunger-induced discontent that, it feared, could trigger communist revolutions in underdeveloped countries. Successive post-World War II administrations made the Food for Peace programme, better known as PL-480, a cornerstone of US foreign assistance. India alone imported nearly 24 million tonnes (mt) of wheat under PL-480 during 1964-66. The US even supplied some 9.1 mt of subsidised wheat and corn to the Soviet Union in 1973, whose ideological significance wasn't lost on anybody. For the US, food shipments, on concessional or outright grant terms, served both as a bulwark against communism and a means to relieve its massive farm surpluses. John F Kennedy estimated that it cost 20 cents a year to store a bushel of wheat and 38 cents to ship the same grain to India. Exporting grain free, then, was cheaper than stocking beyond two years.

The same opportunity — what Kennedy called "turning our great agricultural abundance into a blessing, for ourselves and for all the world" — presents itself, albeit in a smaller manner, to India today. At 90.41 mt as on September 1, the country's public stocks of wheat and rice were the highest ever for this date, with the new paddy arrivals from October only going to add to these. Moreover, the stock pileup has taken place, despite a record 93 mt-plus grain offtake from the Central pool during 2020-21, much of it given out free/near-free post the pandemic. With government agencies procuring over 103 mt last year, the quantities flowing into the Food Corporation of India's warehouses are far in excess of that going out. The "carrying cost" — interest, storage and other expenses — of the excess buffer has been estimated at Rs 5,589 per tonne for 2021-22. Clearly, there is economic as well as diplomatic sense in donating or

bartering this surplus grain abroad. And where better to do it than in our immediate neighbourhood?

Afghanistan is now facing an acute food crisis from a combination of prolonged drought, regime change and associated instability. Sri Lanka, too, is **grappling with** food shortages, worsened by dwindling foreign currency reserves. What stops India from offering, say, 2 mt of wheat to Afghanistan as humanitarian aid and one mt of rice to Sri Lanka against payment in local currency similar to that under PL-480? Wheat can only be eaten or, at worst, diverted as animal feed. There should be no moral compunction, hence, in supplying it even to a regime that India cannot recognise. Grain diplomacy — a simple message that nobody in South Asia shall starve while we are here — can be symbolic of the New India: One that cares and matters to the world.

74. What is the theme upon which the above passage has been framed?
- Food: a powerful diplomacy tool
 - A New India that will not let anyone starve in South Asia
 - The great agricultural abundance
 - The economic aspect of bartering food abroad
 - All of the above
75. How did the food shipments become significant for the US?
- They acted as a defense against communism
 - Helped relieve the US of its massive farm surpluses
 - Created a dominance of the US over the world nations
- Only (ii)
 - Both (i) and (iii)
 - Only (i)
 - All (i), (ii) and (iii)
 - Both (i) and (ii)
76. What is the meaning of the phrase 'grappling with' as used in the passage above?
- Come to grips with
 - Confront
 - Deal with
 - All (a), (b) and (c)
 - Only (b) and (c)
77. In what reference has the line; 'There should be _____ India cannot recognise' been used in the passage above?
- Viz-a-viz providing grains to countries like Afghanistan and Sri Lanka
 - If India fails to provide grains and the US does
 - Stocking of food grains for our own fellow countrymen and not giving to other nations
 - Both (ii) and (iii)
 - None explains the line well
78. Which of the following is true as per the given passage?
- India alone imported nearly 25 million tonnes (mt) of wheat under PL-480
 - It costs 20 cents a year to store a bushel of wheat
 - The "carrying cost" — interest, storage and other expenses — of the excess buffer has been estimated at Rs 5,589 per tonne
 - Both (b) and (c)
 - All are true
79. What is the tone of the passage?
- Satirical
 - Critical
 - Analytical
 - Periphrastic
 - Didactic

Directions (80): The following questions given below carry a situation or a theme based on which you have to identify the option that carries the word that describes the given situation in the best possible manner and mark the same as your answer

80. The students of that batch are mainly focused, stubborn, and determined. They will keep going no matter what and they imagine themselves already achieving what they want, but they don't just stop there. Many people establish goals for themselves. But what

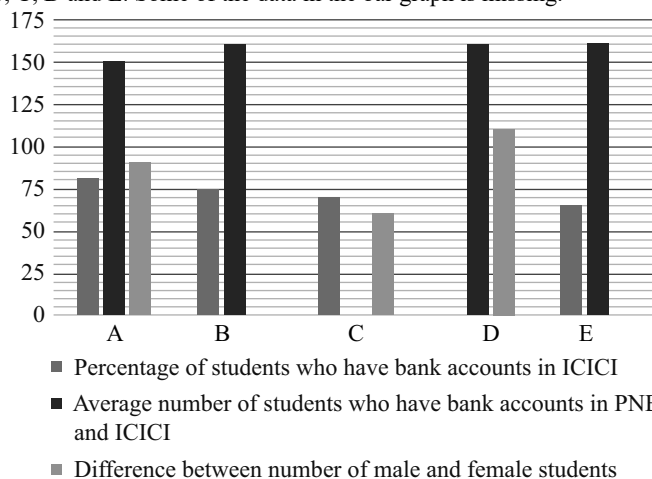
really sets it is that, once the goal them apart is the properly set, next step and to follow it through and make it happen, and to make it, they ask from themselves what is needed in order to reach what they want and divide all the work in several mini-tasks. Then they throw themselves completely and utterly to get each little task to work. And it sounds easy, but takes concentration, hard work, and time and never to give up or be discouraged when things don't go their way. Which word describes the students of this batch?

- Audacious
- Insolent
- Industrious
- Ingenuous
- Naïve

DATA ANALYSIS AND INTERPRETATION

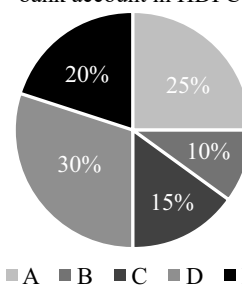
Directions (81-83): Study the following information carefully and answer the related questions.

Following bar-graph represents the percentage of students who have bank accounts in ICICI out of total students in respective college; average number of students who have bank accounts in PNB and ICICI; and difference between number of male and female students in college A, B, C, D and E. Some of the data in the bar graph is missing.



Following pie-chart represents the percentage break-up of number of students who have their bank accounts in HDFC.

% break-up of number of students who have bank account in HDFC



81. If the respective ratio of male students in college A, B and E is 9 : 8 : 5 respectively, then the difference between number of female students in college B and E is how much more/less than the number of students who have bank accounts in HDFC from college C and D together?

- (a) 20 (b) 5 (c) 15
(d) 24 (e) 8

82. Average number of students who have bank accounts in PNB from college B, D and E is 77, then approximately ___% students from college D have bank accounts in ICICI.
(a) 64% (b) 52% (c) 58%
(d) 74% (e) 68%
83. If 226 students from college B and C together do not have accounts in ICICI, then male students in college C are approximately what percent more/less than the number of female students in college A?
(a) 124% (b) 108% (c) 82%
(d) 96% (e) 70%

Directions (84-87): Read the data carefully and answer the following questions.

There is a freelancer who create questions of two difficulty levels, i.e., easy and difficult in five different months.

Note:

I. Pay for each easy question is ₹40 while pay for each difficult question is ₹60.

II. There is deduction in pay for error also.

III. For each error in easy question ₹3 is deducted while for each error in difficult questions ₹5 is deducted.

Table given below shows the total number of questions created by a freelancer, ratio of easy to difficult questions created and average error per question in both type of questions.

Months	Total created questions	Ratio of easy to difficult questions	Average error per question	
			For easy questions	For difficult questions
Jan	60	7 : 5	0.6	0.48
Feb	66	8 : 3	0.75	0.5
Mar	70	4 : 3	0.5	0.6
Apr	69	15 : 8	0.4	0.25
May	66	25 : 8	0.64	0.75

84. What is the total amount received by the freelancer in Jan and Mar together?
(a) ₹6027 (b) ₹5777 (c) ₹5847
(d) ₹6457 (e) ₹5287
85. What is the difference between the total amount received by the freelancer in May for easy questions and total amount received by him in Feb for difficult questions?
(a) ₹847 (b) ₹909 (c) ₹801
(d) ₹869 (e) ₹878
86. If in the month of June, he creates only easy questions which is equal to 20% of the total easy questions created by him in four of the mentioned 5 months (not Feb) and his average error per question is $\frac{8}{17}$, then what is his earning in June?
(a) ₹1507 (b) ₹1416 (c) ₹1312
(d) ₹1108 (e) ₹1627
87. What would be total increment in his earning in April if he did not make errors in the question?
(a) ₹72 (b) ₹84 (c) ₹90
(d) ₹96 (e) ₹108
88. Two persons P and Q started a business with their capitals in the ratio of $(x^2 + 4 - 2x) : (x^2 + 4 + 2x)$ respectively. If ratio of their profit at the end of business becomes $(x^3 + 8) : (x^3 - 8)$ respectively, then find the ratio of their period of investment.
(a) $(x^2 + 4) : (x^2 - 4)$ (b) $(x + 4) : (x - 4)$
(c) $(x^2 + 2) : (x^2 - 2)$ (d) $(x + 2) : (x - 2)$
(e) $(x + 8) : (x - 8)$

89. Find the missing number in the given number series.
91, 103, 87, 112, 71, 137, ?
(a) 37 (b) 35 (c) 33
(d) 36 (e) 39

90. Find the missing number in the given number series.
13, 69, ?, 1714, 8567, 42839, 214192
(a) 342 (b) 324 (c) 242
(d) 256 (e) 343

91. In the following question, read the given statement and compare Quantity I, Quantity II, Quantity III, and Quantity IV on its basis. (Only quantity is to be considered)

Rahul went to a shopping mall and purchased three items A, B and C there. Total labelled price of all the three items together is ₹1500 and items A, B and C are sold at 10%, 20% and 5% discount on their labelled price respectively.

Quantity I: Purchased price of all the three items together for Rahul if purchased price of items A and B together is ₹1000 and effective discount percent on items A and B together is $16\frac{2}{3}\%$.

Quantity II: Labelled price of items A and B together if ratio of purchased price of items A, B, and C for Rahul is 72 : 128 : 57 respectively.

Quantity III: Cost price of all the three items together if ratio of cost price of items A, B and C is 32 : 50 : 25 respectively and labelled price of the items are 25%, 60% and 20% above their cost respectively.

Quantity IV: 1.4 times of sum of difference between labelled price of A and B and difference between labelled price of B and C if purchased price of B for Rahul is ₹355 more than purchased price of C which is ₹75 less than the purchased price of A.

Which of the following is correctly shown in increasing order?

- (a) Quantity III, Quantity II, Quantity IV, Quantity I
(b) Quantity II, Quantity I, Quantity III, Quantity IV
(c) Quantity IV, Quantity I, Quantity II, Quantity III
(d) Quantity III, Quantity IV, Quantity I, Quantity II
(e) Quantity II, Quantity IV, Quantity III, Quantity I
92. In the following question, read the given statement and compare Quantity I, Quantity II, Quantity III, and Quantity IV on its basis. (Only quantity is to be considered)
- Four persons P, Q, R, and S together can do a piece of work in 6 hour. Ratio of efficiency of P to Q is 2 : 3 and time taken by Q alone to do the work is 5 hours more than the time taken by S alone to finish the work. Time taken by persons R and S together to do the work is 12 hours.
- Quantity I:** Time taken by Q and R together to do the work.
Quantity II: Time taken by P and R together to do the work if P works with 80% of his actual efficiency and R works with 240% more than his actual efficiency.
Quantity III: Time taken by person T alone to do the work if persons S and T together can finish that work in 8 hours.
Quantity IV: Time taken to finish the work by P and S if P alone works for first 2 hours after which S alone works for 3rd hour and the same process continues till the work is finished.
- Quantity I __ Quantity II __ Quantity III __ Quantity IV
If @ means >, # means <, \$ means ≥, % means ≤, & means =.
Which of the following symbols will fill the blanks in the same order?
(a) &, @, % (b) @, @, @
(c) #, #, # (d) @, \$, %
(e) @, #, #
93. In the following question, read the given statement and compare Quantity I, Quantity II, Quantity III, and Quantity IV on its basis. (Only quantity is to be considered).
- Ratio of present age of Mayank to that of Manish is 7 : 5 and Manish is 18 years younger to Manoj. Ratio of ages of Manoj to Mukesh

after 12 years will be 5 : 4 respectively and present age of Mukesh is 36 years.

Quantity I: Average present age of all the four persons.

Quantity II: Average present age of the oldest and the youngest person.

Quantity III: Average present age of the second oldest and the second youngest person.

Quantity IV: Present age of Mahesh, if average present age of Mayank, Manish, Manoj, Mukesh, and Mahesh is 36 years.

Quantity = Quantity = Quantity > Quantity
Which of the following options is correct?

- (a) I, II, IV, III (b) II, I, IV, III
(c) III, II, IV, I (d) II, III, IV, I
(e) I, II, III, IV

94. In the question, three statements numbered I, II and III given. You have to decide whether the data provided in the statements are sufficient to answer the question. Read all the statements and give answer.

Find the difference between the interest earned by Arun from scheme A and scheme B.

Statement I: Arun invested ₹60000 in scheme A which offers compound interest at $r\%$ per annum for 3 years.

Statement II: Difference between compound interest and simple interest on ₹40000 at $r\%$ per annum after two years will be ₹256.

Statement III: Arun invested ₹45000 on scheme B which offers simple interest at $r\%$ per annum for 5 years.

- (a) Only I and II (b) Only I and III
(c) Only II and III (d) All I, II and III
(e) Any two of the three

95. The question is followed by three statements. You have to determine whether the data given in the statement(s) is/are sufficient for answering the question.

Find the speed (in m/s) of train A.

Statement I: The difference between the time taken by train A to cross a pole and time taken to cross a bridge of length 60 m at same speed is 2 sec.

Statement II: Length of train B is 50 m more than the length of train A and speed of train B is 30% more than the speed of train A. The ratio of the time taken by train B to cross train A while moving in opposite direction to time taken by train B to cross train A while moving in same direction is 3 : 23.

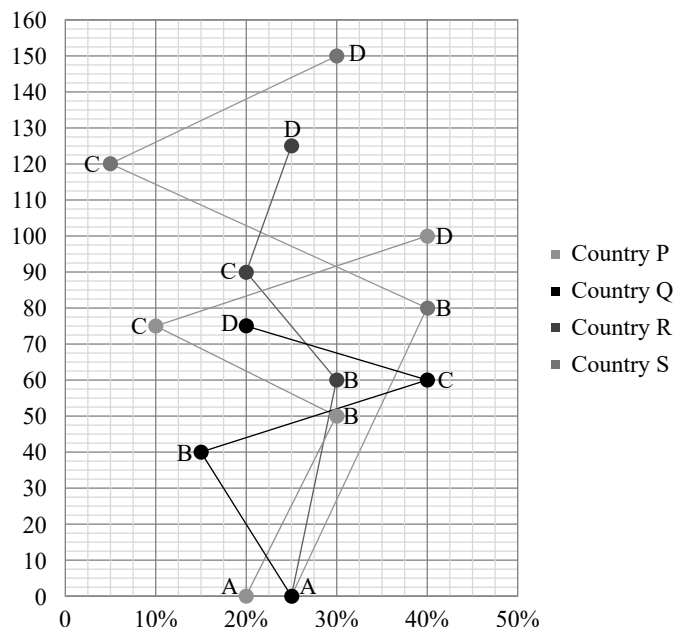
Statement III: Time taken by train A to cross a stationary train C, whose length is 100 m more than train A and time taken by train A to cross a bridge of length 50 m at 15 m/sec less speed are same.

- (a) Only I is sufficient
(b) Only II is sufficient
(c) Only III is sufficient
(d) Either only I or only III is sufficient
(e) Either only II or only III is sufficient

Direction (96-98): Read the data carefully and answer the following questions.

In a sports event, players from four countries P, Q, R and S participate. Players participate in four games A, B, C and D.

In the graph given below, percentage of players participating in each of the four games out of total players from a country is given by horizontal axis value corresponding to the respective game with respect to that country while average number of players (per game) participating in four games from a country is given by the maximum vertical axis value corresponding to a game with respect to that country (i.e. vertical axis value corresponding to D).



96. Out of total players from country P, at least 20 players in game A, at least 30 players in game B, at least 25 players in game C and at least 40 players in game D won 2 medals each. Total number of players who did not win any medal from country P is 120 and remaining won minimum 1 medal each. What is the minimum number of medals won by country P?
(a) 385 (b) 405 (c) 395
(d) 355 (e) 365

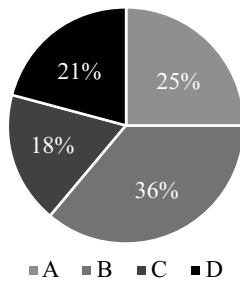
97. From country S, out of total players participated in games A, B, C, and D; 30%, $m\%$, 20%, and $(m + 20)\%$ respectively won the medals. If total medals won by country S is not more than 234, find the range of values of ' m '. (Assume one player won only one medal.)
(a) Less than or equal to 35 (b) More than or equal to 35
(c) Less than or equal to 45 (d) More than or equal to 45
(e) Less than or equal to 55

98. Out of total players participated in game C from four countries, $x\%$ and ' $x + 10\%$ ' are female players from countries P and Q respectively and ' $y\%$ ' and ' $y + 10\%$ ' are female players from countries R and S respectively. Total male players participated in game C from all countries together is 162. Find the values of ' x ' and ' y ' respectively.
(a) 30 and 50 (b) 20 and 60
(c) 10 and 70 (d) Both (a) and (b)
(e) Both (a) and (c)

99. A person covers the distance of 200 km into two parts. He travelled first part in AC car and remaining distance in non-AC car. If the rent of AC car and non-AC car is ₹14 per km and ₹6 per km respectively and he paid total rent of ₹1360, find the distance travelled by him by non-AC car.
(a) 150 km (b) 170 km (c) 190 km
(d) 180 km (e) None of these

Directions (100-103): Study the following information carefully and answer the given questions. The pie chart gives the percentage distribution of flats sold by four dealers A, B, C, and D, out of the total flats sold. The dealers can only sell the ones available with each of them. The dealers deal in only three types of available flats: 1BHK, 2BHK and 3BHK, details of which are given in the table. At least one flat of each type (1BHK, 2BHK and 3BHK) is left unsold with each dealer.

% distribution of flats sold



Dealer	Total no. of 1BHK and 3BHK flats available	Percentage of 2BHK flats available out of the total
A	680	33.33%
B	790	21%
C	650	35%
D	390	50%

Note:

- (i) 65% of the 3BHK flats available with dealer A were not sold.
 - (ii) No 3BHK flat available with dealer B were sold and no 1BHK flat available with dealer C were sold.
 - (iii) 50% of the 2BHK flats available with dealer D were sold.
- 100.** 90% of the total number of 1BHK and 3BHK flats available with D were sold. How many 1BHK and 2BHK flats were sold by A if there were 320 1BHK flats available with A?
 (a) 650 (b) 416 (c) 524
 (d) 560 (e) 472
- 101.** 41% of the 1BHK flats and 70% of the 2BHK flats available with A were sold. The ratio of the number of 2BHK flats and 3BHK flats sold by him was 17 : 7 respectively. What was the total number of flats sold by B?
 (a) 750 (b) 360 (c) 500
 (d) 720 (e) 648
- 102.** 381 flats available with D were not sold. 30% of the 2BHK flats available with B were sold. Which of the following can be the total number of 1BHK flats that were available with B?
 I. 630
 II. 600
 III. 621
 (a) Both I and III (b) Only III
 (c) None of I, II and III (d) Both II and III
 (e) Only I
- 103.** The ratio of the number of 1BHK and 3BHK flats available with C was 4 : 9 respectively. The ratio of the number of 3BHK flats sold by him to the total number of 2BHK flats available with him was 5 : 7 respectively. Which of the following can be the number of flats sold by C that were not 3BHK?
 I. 470
 II. 25
 III. 182
 (a) Only III (b) Both II and III
 (c) All of I, II and III (d) Only II
 (e) Both I and III

Directions (104-106): Study the following information carefully and answer the related questions.

A yearly fest held in three schools A, B and C. In this fest all teachers, all students and students' family members were present. Following

information gives the data regarding number of teachers, students and family members present at the fest.

- Respective ratio of male to female teachers in school A is 2 : 7 while 40% of teachers in school B are females.
- Number of students in school A is 16% less than the number of students in school C.
- Number of male teachers in school C is 4 more than the number of female teachers in same school and equal to 3% of the total number of students in school B.
- 65% of the students in school B are females.
- Number of students in school C is 700 more than the number of students in school B.
- Number of male teachers in school A to C is in the ratio 1 : 3 respectively.
- Respective ratio of number of teachers to number of students in school B is 1 : 16.
- There are 80 teachers in school A and C together which are 50 more than the number of male teachers in school B.

Some additional information is also known:

In school B, only one member from each of 35% of students' family and two members from each of rest students' family were present at the fest. In school C and A together, total 4260 family members were present at the fest. Only one member was present at the fest from each student's family in school A.

- 104.** Respective ratio of female students to female teachers in school A is 25 : 2 and number of male students in school D is 50 more than the number of male students in school A. If 40% students in school D are females, then what is the difference between total number of students in school C and D?
 (a) 180 (b) 100 (c) 150
 (d) 230 (e) 200
- 105.** If the average age of male teachers in school A is 24 years, average age of male teachers in school B is ___ years and the average age of male teachers in school C is 20 years, then the average age of all male teachers in school A, B and C is 21 years.
 (a) 21 (b) 25 (c) 23
 (d) 19 (e) 27
- 106.** In school C, if only one member present at the fest from each of 80% of students' family and $(k - 2)$ members present at the fest from each of rest students' family, then find the value of 'k'.
 (a) 3 (b) 4 (c) 5
 (d) 8 (e) 7

Directions (107-109): Study the data carefully and answer the following questions:

Three boats K, L, and M are traveling some distance in a river, and the speed of the stream of the river is 4 km/h. The speed of the boat L in still water is equal to the speed of the boat K in upstream which is equal to 12 km/h. Boat M travels 35 km in upstream in 2.5 hours and in upstream, Boat L travels 3 km less in upstream than boat M. Boats K and M travel (upstream + downstream) for total 5 hours and 4.5 hours, respectively. Boat L travels 48 km in downstream which is 18 km more than the distance travelled by boat K in downstream

- 107.** What is the ratio of distance covered by boat K in upstream to the distance covered by boat M in downstream?
 (a) 19 : 18 (b) 24 : 25 (c) 32 : 33
 (d) 22 : 23 (e) 21 : 22

108. The time taken by boat *K* in upstream is what percentage more or less than the time taken by boat *L* in upstream?
- (a) 20% more (b) 33.33% less
(c) 18.3% more (d) 12.5% less
(e) 10% less
109. The distance traveled by boat *K* in downstream is how many km more or less than the distance traveled by boat *L* in upstream?
- (a) 1 km more (b) 4 km less
(c) 3 km more (d) 2 km less
(e) 5 km less

Directions (110-113): Study the following information carefully and answer the questions given below.

Some recruitment is done by three companies, i.e., *A*, *B* and *C*. The number of vacancies in company *A*, *B* and *C* are 3, 6 and 1 respectively. The number of candidates, who got interview call per vacancy in company *A*, company *B* and company *C* are 30, $3x$ and $2x$ respectively. The total number of candidates applied in company *C* is one fourth of the total number of candidates applied in company *B* and three fifth of the total number of candidates applied in company *A*. The difference between the number of candidates got interview call in company *B* and *C* is 160.

110. What is the total number of candidates applied in company *B*?
- (a) 450 (b) 600
(c) 300 (d) Can't be determined
(e) None of these
111. The number of candidates who applied in company *C* is what percentage of the number of candidates who applied in company *A*?
- (a) 75% (b) 50% (c) 60%
(d) 80% (e) None of these
112. In company *A*, out of the total number of candidates who got interview call 70% are female and company *A* allotted its posts to male and female in the ratio of 2 : 1 respectively, then find the number of females who did not get the job in company *A*.
- (a) 42 (b) 40 (c) 62
(d) 50 (e) None of these
113. If the difference between the total number of candidates who applied in company *A* and company *C* is 150, find the number of candidates who did not get interview call from company *B*.
- (a) 900 (b) 960 (c) 1050
(d) 720 (e) None of these
114. Three persons *A*, *B* and *C* wrote an examination. The ratio of probability of success to the probability of failure of *A* is 1 : 4. The probability of failure of *B* is $\frac{375}{4}$ % of that of *A* and the probability of success of *C* is $\frac{500}{3}$ % of that of *A*. What is the ratio of the probability that no one will succeed in the examination to the probability that no one will fail in the examination?
- (a) 24 : 1 (b) 22 : 1 (c) 19 : 1
(d) 26 : 1 (e) None of these
115. In an examination P attempted 40 questions so far and 30% of them attempted correct. If P attempted remaining *Z* questions, out of which *Y*% are correct, so in the exam, number of correct and incorrect questions attempted by him are same. If out of *Z* questions attempted, 90% are correct so total correct questions attempted by P

is (*Y* + 24). Find total number of questions in the exam. Consider P attempted all questions in the exam.

- (a) 240 (b) 150 (c) 120
(d) 170 (e) None of these

GENERAL, ECONOMY/BANKING AWARENESS

116. In September 2022, the Monetary Policy Committee of the Reserve Bank of India has decided to increase Bank Rate to ____.
- (a) 6.0 (b) 6.15 (c) 6.25
(d) 6.5 (e) None of these
117. Minicoy, Thundi Beach and Kadmat Beach, are the proud entrants in the coveted list of Blue Beaches, an eco-label given to the cleanest beaches in the world. These two beaches are located in which of the following Union Territory?
- (a) Lakshadweep (b) Puducherry
(c) Nicobar (d) Andaman
(e) Daman and Diu
118. What is the name of the web-based application which has been launched by RBI Governor Shaktikanta Das on 6th Oct 2022, to make the supervisory processes more robust?
- (a) YOGYA (b) URJA
(c) DAKSH (d) PRAKHAR
(e) PRAWAH
119. Tennis player Carlos Alcaraz won the title match of US Open 2022, he defeated Casper Ruud. Casper Ruud belongs to which of the following countries?
- (a) USA (b) England (c) Norway
(d) Greece (e) Spain
120. Rishi Sunak became the _____ UK Prime Minister on October 25, after meeting King Charles III at Buckingham Palace.
- (a) 54th (b) 55th (c) 56th
(d) 57th (e) 58th
121. PM Narendra Modi inaugurated the first greenfield airport, Donyi Polo Airport, in which of the following states?
- (a) Arunachal Pradesh (b) Assam
(c) West Bengal (d) Mizoram
(e) Nagaland
122. India ranked 132 out of 191 countries in the 2022 Human Development Index (HDI), according to a report released by the United Nations Development Programme (UNDP). Who among the following was on top as per the HDI Index 2022?
- (a) Nepal (b) Bhutan
(c) Switzerland (d) Bangladesh
(e) Pakistan
123. Which city has been declared the first 'cultural and tourism capital' of Shanghai Cooperation Organization (SCO) in July 2022?
- (a) Varanasi (b) Beijing (c) Jakarta
(d) Singapore (e) Mumbai
124. Prime Minister Shri Narendra Modi laid the foundation stone of the C-295 transport aircraft manufacturing facility - the country's first in the private sector—in Vadodara, Gujarat. The C-295 aircraft has been manufactured by Airbus Defence and Space S.A., Spain, and which of the following Indian companies?

- (a) Electronics Corporation of India Ltd
(b) TATA Advanced Systems Limited
(c) BAE Systems India Services Pvt Ltd
(d) Boeing India Pvt Ltd
(e) Hindustan Aeronautics Limited (HAL)
125. The Nobel Prize in Economics 2022 has been given to US economists Ben S. Bernanke, Douglas Diamond, and Philip H. Dybvig. What was the topic of their research?
(a) On the reduction of the risk of financial crises developing into short-term boom.
(b) On the reduction of the risk of financial crises developing into long-term depressions.
(c) On the reduction of the risk of medical crises developing into long-term depressions.
(d) On the reduction of the risk of financial crises developing into short-term depressions.
(e) On the reduction of the risk of medical crises developing into long-term boom.
126. 14th World Spice Congress (WSC), organized by Spices Board of India in association with various trade and export forums, was scheduled to be held in which of the following cities in February 2023?
(a) Navi Mumbai (b) New Delhi
(c) Surat (d) Trivandrum
(e) Kochi
127. In November the Reserve Bank of India (RBI) RBI imposed a monetary penalty of how many rupees on Vakrangee Ltd. for non-compliance with certain provisions related to White Label ATM (WLA) directions?
(a) Rs. 1.5 crores (b) Rs. 1.76 crores
(c) Rs. 1.75 crores (d) Rs. 1.67 crores
(e) Rs. 1.78 crores
128. Which Bank has received the RBI's approval to open a special vostro account with Gazprombank of Russia for trade settlement in Indian rupees?
(a) UCO bank (b) SBI Bank
(c) HDFC Bank (d) ICICI Bank
(e) Indusind Bank
129. In April 2022, SEBI reduced the time taken for the listing of public issue units of REIT and InvIT to _____ from the present 12 days.
(a) 2 days (b) 4 days (c) 5 days
(d) 6 days (e) 10 days
130. The world-famous International Sand Art Festival will take place in December 2022 in Odisha at the Chandrabhaga Beach, which is located near to _____.
(a) Padamapur (b) Batighara
(c) Balukhand (d) Konark
(e) Puri
131. Which of the following mutual fund companies has launched India's first Auto Index Fund in September 2022?
(a) SBI Mutual Fund
(b) HDFC Mutual Fund
(c) ICICI Prudential Mutual Fund
(d) Navi Mutual Fund
(e) None of the above
132. Which of the following Indian banks has become the first to issue an e- Bank Guarantee?
(a) Union Bank (b) ICICI Bank
(c) SBI Bank (d) Axis Bank Bank
(e) HDFC Bank
133. The lemon and lime-flavored soft drink 'Sprite' has become a billion-dollar brand in the Indian market. 'Sprite' is owned by which of the companies?
(a) Parle Agro
(b) Pepsico Inc.
(c) Bisleri International private limited
(d) The Coca-Cola Company
(e) None of these
134. Which of the following will be the 12th company to get the status of the 'Maharatna' company?
(a) Rural Electrification Corporation Limited
(b) Power Finance Corporation Limited
(c) Hindustan Aeronautics Limited
(d) Container Corporation of India Limited
(e) Coal India Limited
135. Which of the following states has passed the bill to ban online gambling and to regulate online games keeping in view the well-being of citizens?
(a) New Delhi (b) Tamil Nadu
(c) Uttar Pradesh (d) New Delhi
(e) Karnataka
136. The Competition Commission of India (CCI) approved the merger between Sony Pictures Networks India (SPN) and which of the following Indian media networks in October 2022?
(a) 9X Media India
(b) The Walt Disney Company India
(c) Viacom 18 Media Private Limited
(d) Zee Entertainment Enterprises Ltd
(e) DD Network India
137. Which of the following is true about the Social Stock Exchange?
(A) The idea of SSE was first floated by Finance Minister Nirmala Sitharaman in her Budget speech for the financial year 2021-2022.
(B) The idea of the Social Stock Exchange (SSE) is to provide social enterprises with an additional avenue to raise funds.
(C) SEBI has asked social enterprises raising funds using SSE to disclose the Annual Impact Report (AIR) within 60 days from the end of the financial year.
(a) A, B & C (b) A & C only
(c) B & C only (d) B only
(e) C only
138. The Prime Minister launched the National Logistics Policy. According to the policy, India aims to be among the top 25 countries in the Logistics Performance Index (LPI) ranking by which year?
(a) 2024 (b) 2028 (c) 2030
(d) 2032 (e) 2035
139. As per Reserve Bank of India (RBI) guidelines, Commercial Papers (CPs) are issued in a minimum denomination of _____.
(a) Rs. 1 lakh (b) Rs. 2 lakh
(c) Rs. 3 lakh (d) Rs. 5 lakh
(e) Rs. 50,000
140. In October 2022, the Genetic Engineering Appraisal Committee (GEAC) approved the commercial cultivation of genetically modified seeds of which of the following?

- (a) Mustard (b) Sesame
(c) Soya (d) Canola
(e) Sunflower
141. As per the Union Budget 2022-2023, the Productivity Linked Incentive (PLI) scheme in 14 sectors has targeted to create 60 lakh new jobs and an additional production of Rs. 30 lakh crore during next 5 years. PLI scheme was launched by PM Modi on 11 November 2020 in how many key sectors?
(a) 8 sectors (b) 9 sectors
(c) 10 sectors (d) 11 sectors
(e) 12 sectors
142. Who among the following can not open the savings account in the Scheduled Commercial bank?
A. Municipal Corporations/Committees
B. Panchayat Samities
C. Metropolitan Development Authorities
D. State Government
(a) Only A (b) C and D
(c) B and C (d) A, B, and C
(e) Only D
143. Who among the following was appointed as the Chief Information Commissioner (CIC) in March 2020?
(a) Ajay Bhushan Pandey (b) Bimal Julka
(c) Amita Pandove (d) Satish Kumar Sharma
(e) None of these
144. Which bank has entered into an agreement to buy a 2 percent stake in asset reconstruction company NARCL for Rs 55 crore in March 2022?
(a) State Bank of India (b) Punjab National Bank
(c) Punjab & Sind Bank (d) Reserve Bank of India
(e) HDFC Bank
145. The Union Minister for Finance & Corporate Affairs, Smt. Nirmala Sitharaman, tabled the Union Budget 2022–23. Consider the following statements about the Union Budget.
A. India's economic growth is estimated at 9.2% to be the highest among all large economies.
B. 1600 crore outlay for implementation of the Ken – Betwa link project.
C. Udyam, e-shram, NCS and ASEEM portals to be interlinked.
(a) Only A (b) B and C (c) A and C
(d) Only C (e) Only B
146. The Reserve Bank Of India (RBI) has released a concept note on the CBDC. It is a digital form of currency notes issued by the central bank. Both the C in the acronym stands for _____ and _____ respectively.
(a) Central, Currency (b) Currency, Central
(c) Credit, Central (d) Credit, Currency
(e) Central, Credit
147. Indian rocket startup _____ has raised \$51 million in a funding round led by Singapore sovereign investor GIC, which it plans to use to begin commercial satellite launches by the middle of next year.
(a) Skyroot Aerospace (b) Aadyah Aerospace
(c) Asteria Aerospace (d) Hampson Industries
(e) Newtech Aerospace
148. _____ and Indian Banks' Association (IBA) are jointly organising FIBAC – our annual banking conference on November 2 and 3 in Mumbai.
(a) State Bank of India (SBI)
(b) Reserve Bank of India (RBI)
(c) Confederation of Indian Industry (CII)
(d) Federation of Indian Chambers of Commerce and Industry (FICCI)
(e) Associated Chambers of Commerce and Industry of India (ASSOCHAM)
149. On October 19, 2022, the 'Mission Schools of Excellence' project was launched in Gandhinagar, Gujarat. Who among the following has launched this project?
(a) President Droupadi Murmu
(b) PM Narendra Modi
(c) Vice President Jagdeep Dhankar
(d) Home Minister Amit Shah
(e) S. Jayshankar
150. SIDBI partnered with _____ co-operative bank to empower MSMEs.
(a) Saraswat (b) Shivalik Mercantile
(c) Abhyudaya (d) SVC
(e) Bharat
151. The Government of India has announced to issue sovereign green bonds of how much in the first half of FY23?
(a) \$3.3 Billion (b) \$3.4 Billion
(c) \$3.5 Billion (d) \$3.6 Billion
(e) \$3.7 Billion
152. The RBI allowed large modern currency chests to increase the service charges on cash deposited by non-chest bank branches from the existing rate to a maximum of ₹ _____ per packet
(a) 5 (b) 8 (c) 11
(d) 14 (e) 17
153. Which bank recently won the global 'Celent Model Bank' award for building a best-in-class Enterprise Payments Hub (EPH)?
(a) IndusInd Bank (b) Kotak Mahindra Bank
(c) ICICI Bank (d) Axis Bank
(e) HDFC Bank
154. The Reserve Bank of India (RBI) has imposed a penalty of Rs 57.50 lakh on _____ for non-compliance with the directions.
(a) Kotak Mahindra Bank
(b) Indian Overseas Bank
(c) Axis Bank
(d) State Bank of India
(e) Union Bank of India
155. The Reserve Bank of India (RBI) has cancelled the licence of _____ citing inadequate capital & earning prospects in September 2022.
(a) Jila Sahakari Kendriya Bank
(b) The Laxmi Co-operative Bank Ltd
(c) Goa State Co-operative Bank
(d) Warud Urban Co-operative Bank
(e) None of the above

Answer Key & Solutions

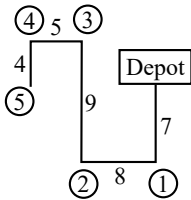
Solutions (1-3):

Month	Date	Person	Fruits
Jan.	11/13	Banu	Apple
	30	Hari	Strawberry
Apr.	11	Cathy	Mango
	28	Jeni	Grapes
Jun.	13	Fath	Banana
	16	Dani	Kiwi
Aug.	01	Anu	Papaya
	18	Imu	Guava
Dec.	11/13	Emi	Plum
	30	Guna	Pear

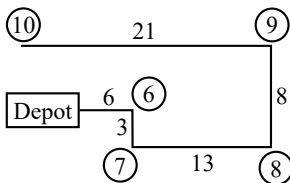
- (b) If Kiwi and Plum interchange their position, then Anu was born immediately after the one who likes Plum.
- (b) Only II is true.
- (d) Both I and III is the answer.

Solutions (4-6):

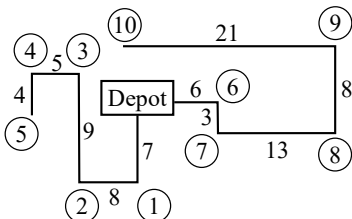
For bus P:



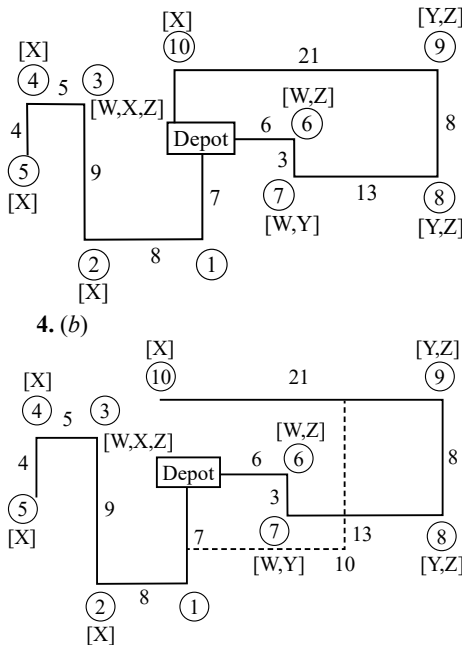
For bus Q:



Now combining the two images we get,



'Z' is assigned to the stoppage if distance between the stoppage and depot is even number which means if we want to calculate the distance between depot and stoppage 2, we would get $(8 + 7) = 15$ m which is not even number. Again, the distance between depot and stoppage 3 is $(8 + 7 + 9) = 24$ m which is even number. Therefore, stoppage 3 would be named as Z.



Hence the distance is $= (6 + 13 - 10) = 9$ meters.

- (d) The correct answer is Stoppage 1 - Stoppage 10 as except this every pair shows inclination from south to east.

- (a) Stoppage 3 assigned maximum names.

Solutions (7-10):

Persons = 21

6 posts in decreasing order - ED, DGM, AGM, Manager, AM and Clerk.

On any post, the number of persons posted is one more than the number of person posted in the immediate senior post. It means there are 6 Clerks.

5 AMs, 4 Managers, 3 AGMs, 2 DGMs and 1 ED.

Posts	Persons
ED (1)	G
DGM (2)	S, F
AGM (3)	C, Q, R
Manager (4)	T, N, B, U
AM (5)	A, E, H, D, V
Clerk (6)	P, L, M, W, X, O

- No transfer happens for the person who belongs to the two highest positions and they become leaders in legal, Marketing and HR respectively according to the alphabetical order. → The two highest positions are ED and DGM. The persons on these posts in alphabetical order are F, G and S. So, the leader of the legal department is F, Marketing is G and HR is S.
- Persons whose name starts with a letter that comes before M in alphabetical order and he/she works with a person whose

name starts with a vowel went to HR (excluding the vowels). → Such persons are L (with O), H, D with (with A and E), B (with U).

- Persons whose name starts with a letter which comes after M in alphabetical order and he/she works at a post which has even number of persons went to Marketing. → Posts having even number of persons are Manager and Clerk. Among them, T, N, U, P, W, X, O went to Marketing.
- Rest will go to the Legal department. → They are C, Q, R, A, E, V and M.

7. (d) all except F are AMs.

8. (e) B who is the manager is transferred to the HR department.

9. (b) C is not the manager.

10. (b) U is transferred to the marketing department.

Solutions (11-13):

For the first symbol - if the given word starts with vowel then write A and if the word starts with consonant then write B.

For the second symbol-

If total number of alphabets is 3 then use symbol - \$

If total number of alphabets are 4 use symbol - &

If total number of alphabets are 5 use symbol - @

If total number of alphabets are 6 use symbol - %

If total number of alphabets are 7 use symbol - ^

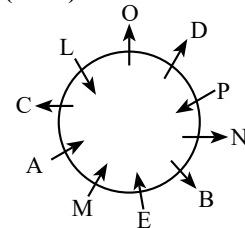
11. (d) The code of "SCHOOL" is "B %".

12. (c) The code of 'America Africa' is 'A ^ A %'.

13. (a) The code for 'Rain Starts Even' is 'B& B% A&'.

14. (e) This could weaken the statement if the timing of the bulk order is just before or during winter. If demand drops seasonally, the huge order might be a poor decision.

Solutions (15-16):



15. (b) P sits fourth to the right of C.

16. (b) M sits opposite to D.

Solutions (17-20):

Floor	Person	Language
9	F	Assamese
8	C	Bengali
7	H	Gujrati
6	B	Odia
5	A	Sanskrit

4	E	English
3	G	Telegu
2	D	Hindi
1	J	Kannada

17. (e) B lives immediately above the one, who can speak Sanskrit.
 18. (e) 9th floor - F - Assamese is the correct combination.
 19. (d) Two persons live between A and C.
 20. (a) H speaks Gujarati.
 21. (e) Statement I:

Holidays are fun → pnj [kol] sip

(Winter) Holidays coming → [kol] (bit) pop

(Winter) is cold → (bit) fog sap

Word	Code
Holidays	Kol
Winter	Bit
Coming	Pop

Statement I is alone sufficient to answer the question.

Statement II:

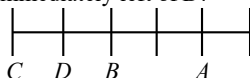
They are coming → pop top gig

How are they → top sun gig

The code for 'They' and 'Are' are 'top' and 'gig'.

Therefore, the code for 'coming' is 'pop'. Statement II is alone sufficient to answer the question.

22. (d) Statement I: A sits second from one end. One person sits between A and B. C sits immediately left of D.

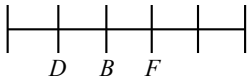


F sits to the right of C.

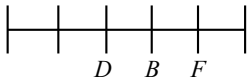
Hence, Statement I alone is not sufficient to answer the question.

Statement II: B sits third from one of the ends. F who does not sit at the corner is one of the positions right to B. The person sits between D and F.

Case I:

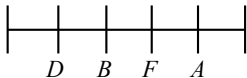


Case II:

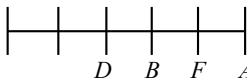


Two people sit between D and A.

Case I:

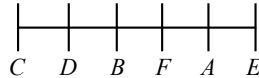


Case II:



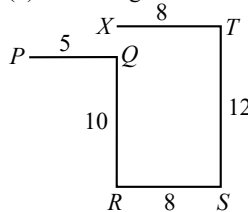
Hence, Statement II alone is not sufficient to answer the question.

If we combine the both statements we get,



Therefore, both statements are necessary to answer the question.

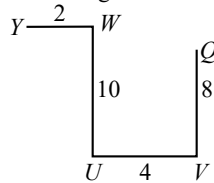
23. (a) According to statement I:



Therefore, X is in the north direction with respect to Q.

Hence, Statement I is alone sufficient to answer the question.

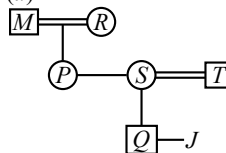
According to statement II:



Here, we only know the distance between X and Y but the direction is not mentioned. So, we cannot draw point X in statement II.

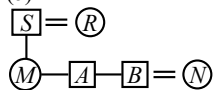
Hence, Statement II alone is not sufficient to answer the question.

24. (d)



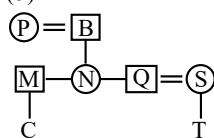
When we use then sign '&' then J is Son of T and when * is used then in that case J is daughter of T.

25. (e)



Thus, N is the sister-in-law of M.

26. (c)



27. (b) I. Children should play outdoor games to keep their body and mind fit and fine – This action is true as outdoor games keep one's healthy and prevent many health problems like obesity, muscular strain etc. II. Parents should stop using mobile so that their children could not play mobile games – This action is false as parents should prohibit their children from playing mobile games but stopping using mobile is not an option.

III. Parents should tell their children to focus only on study not any kind of games – This action is false as playing is also important for children. It helps in the development of children's mind.

Thus, only I follows.

28. (d) (A) Cognizant pays its employees even when they are not on billable projects - This is the assumption behind the decision taken by Cognizant. This is what bench time is all about.

(B) More layoffs will take place in Cognizant in the coming days - This is not the assumption rather it is the effect of the decision taken by Cognizant.

(C) Headcounts in Cognizant are definitely going to shrink in the days to come globally - This is also not the assumption just a definite statement about the future.

(D) The company is trying to reduce its overhead expenses incurred on the employees - This is also not the assumption as many employees can also be on billable projects.

(E) Pay and perk is set to decrease for Cognizant employees in the coming years - The passage is talking about bench time only and not anything about increased cost or decreased profit that can result in the reduction of pay and perk. So, this is also not the assumption.

29. (a) (a) Cognizant pays its employees even when they are not on billable projects - This is the assumption behind the decision taken by Cognizant not the effect. This is what bench time is all about.

(b) More layoffs will take place in Cognizant in the coming days - This will be the effect of the decision taken by Cognizant.

(c) Headcounts in Cognizant are definitely going to shrink in the days to come globally. This can not be effect as its say that headcounts will reduce soon but reduction of headcounts may take months.

(d) The company is trying to reduce its overhead expenses incurred on the employees - This is also not the effect as it says about possible intensive of the company.

(e) Pay and perk is set to decrease for Cognizant employees in the coming years - The passage is talking about bench time only and not anything about increased cost or decreased profit that can result in the reduction of pay and perk. So, this is also not an effect.

Solutions (30-33):

States	Persons
Tripura	P(53) U(51), R(29)

States	Persons
Manipur	S(92), V(54), W(61) T(64)
Assam	Q(46), X(18)

30. (b) The age of V is 54 years.
31. (e) X belongs to Assam.
32. (d) Age $(R + S) = 29 + 92 = 121$
33. (c) Among the given options, all except U belong to Manipur.
34. (b) Given Statement: $S @ T \leq Q + R$; $Q * P @ W$
 On converting: $S < T > Q \geq R$; $Q = P < W$
 On combining: $S < T > Q = P \geq R$, $S < T > Q = P < W$, $R \leq Q = P < W$
Conclusions:
 I. $R \# W \rightarrow R \leq W \rightarrow$ False (as $R \leq Q = P < W$, therefore $R < W$)
 II. $W \leq R \rightarrow W > R \rightarrow$ True (as $R \leq Q = P < W$, therefore $R < W$).
 Hence, only conclusions II follows.
35. (d) Given Statement: $S @ T \leq Q + R + U$; $Q * P \# W \# V$
 On converting: $S < T > Q \geq R \geq U$; $Q = P \leq W \leq V$
Conclusions:
 I. $V \leq U \rightarrow V > U \rightarrow$ False (as $U \leq R \leq Q = P \leq W \leq V$, therefore $U \leq V$)
 II. $U \# V \rightarrow U \leq V \rightarrow$ True (as $U \leq R \leq Q = P \leq W \leq V$, therefore $U \leq V$)
 III. $R @ T \rightarrow R < T \rightarrow$ True (as $T > Q = P \geq R$, therefore $R < T$)
 Hence, only conclusions II and III follows.
36. (b) Give statement: $U @ V + W \# Y * X$; $V + P \leq Q$
 On converting: $U < V \geq W \leq Y = X$; $V \geq P > Q$
 (A) $P \# X \rightarrow P \leq X \rightarrow$ False (as $P \leq V \geq W \leq Y = X$, no relation)
 (B) $Q @ U \rightarrow Q < U \rightarrow$ False (as $Q < P \geq V > U$, no relation)
 (C) $W \# X \rightarrow W \leq X \rightarrow$ True (as $W \leq Y = X$, $\therefore W \leq X$)
37. (c) (a) The number of people affected by water-borne diseases in City X during this rainy season and not over the past years. So, this can't be inferred from the passage.
 (b) The hospitals strive to help people improve their productivity and reduce risks of diseases and injury in a cost-effective way. But we cannot conclude that the people taking treatment are poor or rich. So, this cannot be inferred.
 (c) The government has opened four new civil hospitals in City X in the beginning

of the year to treat this disease by providing better Health Services. These hospitals strive to help people improve their productivity and reduce risks of diseases and injury in a cost-effective way. So, this can be inferred.

(d) The hospital help people improve their productivity and reduce risks of diseases and injury in a cost-effective way. From this we cannot conclude that before this people had spent more money for their treatment. So, this cannot be inferred.

(e) Drinking contaminated water cause water borne diseases including diarrheal disease. This is true but this is not the main inference to be taken.

38. (c) The passage is talking about the Smart city scheme that will be implemented by the government. So anything related to rural areas is out of scope; further creation of jobs in the rural sector has not been mentioned in the given passage.

The passage does not specify the activities planned by the government under the smart city scheme. Thus, options (a) and (d) will get eliminated.

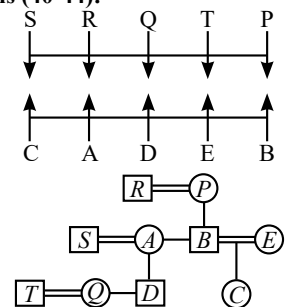
Again, option (b) includes the impact of the scheme in the rural areas which is beyond the scope of the scheme.

This statement "The decision to make cities smart will be taken not by governments but by the people of the city" indicates that the scheme will involve citizen participation in some respect. Thus, enhanced citizen and government engagement is a possible effect of the scheme.

Also, the scheme is meant for cities and effective management of urbanization. Therefore, one of the effects would be the implementation of reforms in urban governance.

39. (c) According to the passage above, the policy that China has taken into consideration for e.g. lock downs to contain the pandemic and cracking down on high debt companies illustrates the reason that why China is slowing down faster than the world. Therefore statement III can be clearly inferred from the passage. Moreover, if the countries started decoupling with China five years ago implies that sometime back they were in close sync with China, hence statement I can also be inferred from the passage. Statement II however cannot be logically derived from the passage and therefore is a fact which needs more evidence to be inferred.

Solutions (40-44):



40. (c) C is cousin of Q.
41. (b) All the persons sit at corner except R.
42. (e) T is the son-in-law of A.
43. (d) D sits opposite to the one who sits immediate left of R.
44. (b) P is the mother-in-law of E.
45. (e) I. CONFIDENT

WORD	C (3)	O (15)	N (14)	F (6)	I (9)	D (4)	E (5)	N (14)	T (20)
	-3	-3	-3	-3	-3	-3	-3	-3	-3
After Change	Z (26)	L (12)	K (11)	C (3)	F (6)	A (1)	B (2)	K (11)	Q (17)
A-Z	A	B	C	F	K	K	L	Q	Z

II. VOCAB

WORD	V (22)	O (15)	C (3)	A (1)	B (2)
	-3	-3	-3	-3	-3
After Change	S (19)	L (12)	Z (26)	X (24)	Y (25)
A-Z	L	S	X	Y	Z

III. EXERCISE

WORD	E (5)	X (24)	E (5)	R (18)	C (3)	I (9)	S (19)	E (5)
	-3	-3	-3	-3	-3	-3	-3	-3
After Change	B (2)	U (21)	B (2)	O (15)	Z (26)	F (6)	P (16)	B (2)
A-Z	B	B	B	F	O	P	U	Z

Therefore, none words end with a vowel.

46. (b) 'creation of a vibrant market for corporate bonds' can fill in the blank and carry forward the given sentence in the most logical way so as to make the sentence coherent and contextually correct. Hence, option (b) is the right answer choice.
47. (d) 'against the loan to protect depositors' can fill in the blank and carry forward the given sentence in the most logical way so as to make the sentence coherent and contextually correct. Hence, option (d) is the right answer choice.
48. (e) 'clientele besides a growing depositor base' can fill in the blank and carry forward the given sentence in the most logical way so as to make the sentence coherent and contextually correct. Hence, option (e) is the right answer choice.
49. (c) 'to fix their own remuneration plans' can fill in the blank and carry forward the given sentence in the most logical way so as to make the sentence coherent and contextually correct. Hence, option (c) is the right answer choice.
50. (b) The correct phrase in the given sentence is "is welcome reform that would shore up transparent price."
51. (c) The correct phrase in the given sentence is "pay tax on their purchases directly to the government."
52. (d) The correct phrase in the given sentence is "enriching incentives can be perceived as alarming disincentives."
53. (b) 'Upheld' means to confirm or support, and 'relinquish' means to voluntarily cease to keep or claim; give up. 'Abdicate' means to renounce one's throne, which is a synonym in the context of giving up or renouncing something. Thus, option (b) indicates the synonyms of the highlighted words.
54. (d) 'Provoke' means to stimulate or incite, and 'condemn' means to express complete disapproval of. 'Appease' means to assuage or satisfy, while 'acclaim' means to praise enthusiastically and publicly. Thus, option (d) indicates the antonyms of the highlighted words.
55. (c) 'Rapid' means happening in a short time or at a great rate, and 'expansion' means the action of becoming larger or more extensive. 'Unhurried' is an antonym of 'rapid', and 'reduction' is an antonym of 'expansion'. Thus, option (c) indicates the antonyms of the highlighted words.
56. (c) In the context of declining revenues of traditional journalistic platforms and interventions against digital news networks, the word that best fits is "plummeting," which means falling or dropping steeply.
57. (b) The phrase "receptiveness of the media fraternity before the powers that be" suggests that the media fraternity is yielding or surrendering to the powers. Thus, "capitulation," meaning ceasing to resist, fits the context.
58. (a) The phrase "the ensconce has turned out to be a rather stubborn species" suggests that journalism has persisted despite challenges. "Vocation," meaning a type of work or calling, fits the context.
59. (d) The sentence talks about journalism being esteemed or respected even in the face of threats. "Resurrected," meaning revived or revitalized, fits the context.
60. (c) The sentence discusses the dangers faced by journalists, including intimidation and threats of violence. "Incarceration," meaning imprisonment, fits the context.
61. (c) The sentences given can be arranged in the sequence of "a-d-b" as the coherent paragraph as it states about practicing self care to improve mental health. However, sentence (3) states about the parenting role in managing child's mental health. Therefore, it fails to find relevance with the other sentences. (e) None of these.
62. (a) The sentences given can be arranged in the sequence of "c-d-b" as the coherent paragraph as it is convincing about eating right the connection between gut health and mental health. However, sentence (1) states about the eating disorder among teenagers. Therefore, it fails to find relevance with the other sentences.
63. (b) The sentences given can be arranged in the sequence of "d-c-a" as the coherent paragraph as it states about NASA delays Psyche mission. However, sentence (2) states about the data science tools. Therefore, it fails to find relevance with the other sentences.
64. (b) The error lies in sentence 3. The verb 'attributed' means 'regard something as being caused by.' It takes the preposition 'to' after it. Thus, the correct sentence is: One major reason for the Soviet Union suffering an economic bust in 1991 was **attributed** to the Kremlin.
65. (a) The error lies in sentence A. In this sentence, the subject is singular '**political problem**'. It should be followed by a singular verb i.e. '**is**'. Thus, the correct sentence is: Another political problem that has contributed to the melting of Arctic ice sheets is the improved navigability of the northern oceans.
66. (d) "Your brain works with chemicals, with neurotransmitters like dopamine and serotonin. Our materials are able to interact electrochemically with them," says Alberto Salleo, a materials scientist at Stanford University who wrote about the potential for organic neuromorphic devices in the 2021 Annual Review of Materials Research.
67. (a) According to the passage: In effect, each neuron serves as both a calculating device integrating the value of all the signals it has received and a memory device: storing the value of all of those combined signals as an infinitely variable analog value, rather than the zero-or-one of digital computers. Integrating means 'Combining or amalgamating the value'. Therefore, the word 'integration' is a suitable word to fill the blank as given in the passage. All other words are not contextually correct to fit in the blank.
68. (e) Both Statements (b) and (c) are correct. It describes the increasing interest in neuromorphic computer circuits that mimic how human neural connections, or synapses, work. These statements also underscores Conventional digital computers work one step at a time, and their architecture creates a fundamental division between calculation and memory. On the other hand, statement (a) is out of the context.
69. (b) The passage underscores that flexible organic circuits that mimic biological neurons could increase processing speed and might someday hook right into your head. The appropriate title of the passage is 'Developing computer chips which act more like brain cells'.
70. (e) According to the given passage 'Researchers have developed a number of different memristive devices that mimic this ability'. A memristor is a non-linear two-terminal electrical component relating electric charge and magnetic flux linkage. Therefore, memristive is a name of device and a unique name which can not be replaced with the other word.
71. (d) According to the passage, 'the value of all the signals combined is effectively stored by each neuron as an infinitely changeable analogue value rather than the zero-or-one of digital computers, acting as both a memory device and a computing device that integrates the value of all the signals it has received.' As all the statements are correct and explain the functioning of the brain in effect with each neuron.
72. (c) According to the passage, "When a voltage is applied, electric current drives positively charged ions from one layer into the other. changing how easily the second layer will conduct electricity the next time it is exposed to an electric current. It's a way of letting physics do the computing", says Matthew Marinella, a computer engineer at Arizona State University in Tempe who researches neuromorphic computing. The technique also liberates the computer from strictly binary values. Hence the correct answer choice is option (c).
73. (a) The correct statement is: As the cells produce dopamine, a nerve-signaling chemical, the dopamine changes the electrical state of the artificial half of the device.

The positions of words in the given sentence are incorrect. Hence swapping the words in their correct positions will frame a grammatically correct and contextually meaningful sentence. The word (1) will be swapped with word (4) and word (2) will be swapped with the word (3) to make the sentence meaningfully correct. Hence, the correct option choice is (a).

74. (b) **Theme of the passage** → is the message that the author wants to convey to the readers

Title of the passage → A title does not only cover the core idea (the content) of the passage, but also expresses the author's tone.

75. (e) Point (iii) is nowhere being talked about and hence can be eliminated. Referring to the following lines from the first paragraph can be referred to for the answer: For the US, food shipments, on concessional or outright grant terms, served both as a bulwark against communism and a means to relieve its massive farm surpluses.
76. (d) To grapple with means to face or deal with a very severe issue or a situation. All the options given in (1), (2) and (3) carry the same meaning and hence explain the asked phrasal verb properly.
77. (a) The given line is trying to explain that providing food grains for free to even those countries that we don't keep in a positive light (like Afghanistan) will not make us feel guilty as what would a country misuse the food grains for...
78. (d) Option (a) is incorrect as; the correct quantity imported by India was 24 mt and not 25 mt. The reference can be found in the first few lines of the first paragraph. The other two points are mentioned as it is in the given passage.
79. (c) The author has tried to provide reasons, logic, or justification to support an idea. Some facts have also been used to explain the author's perspective about how the idea of New India can create an impact. Such a passage is said to carry an 'Analytical' tone.
80. (c) Of the given options, we understand that a word that means Hard working, laborious and determined will be the best fit for the answer. That word out of the options is 'industrious'.

Solutions (81-83):

Let number of students who have bank accounts in HDFC from all given colleges together = n
Then, $2 \times 35 = 15\%$ of $n + 20\%$ of n

$$\Rightarrow 70 = 35\% \text{ of } n$$

$$\Rightarrow n = 200$$

Then, number of students who have bank accounts in HDFC from college:

$$A = 25\% \text{ of } 200 = 50$$

$$B = 10\% \text{ of } 200 = 20$$

$$C = 15\% \text{ of } 200 = 30$$

$$D = 30\% \text{ of } 200 = 60$$

$$E = 20\% \text{ of } 200 = 40$$

From the graph we have:

College	Percentage of students who have bank accounts in ICICI	Average number of students who have bank accounts in PNB and ICICI	Difference between number of male and female students
A	80%	155	90
B	75%	160	
C	70%		60
D		165	110
E	65%	160	

Total number of students who have bank accounts in PNB and ICICI in college A

$$= 2 \times 155 = 310$$

Total number of students who have bank accounts in PNB and ICICI in college B

$$= 2 \times 160 = 320$$

Total number of students who have bank accounts in PNB and ICICI in college D

$$= 2 \times 165 = 330$$

Total number of students who have bank accounts in PNB and ICICI in college E

$$= 2 \times 160 = 320$$

$$\text{Total students in college A} = 310 + 50 = 360$$

$$\text{Male students in college A} = \frac{(360 + 90)}{2} = 225$$

$$\text{Female students in college A} = 225 - 90 = 135$$

$$\text{Total students in college B} = 320 + 20 = 340$$

$$\text{Total students in college D} = 330 + 60 = 390$$

$$\text{Male students in college D} = \frac{(390 + 110)}{2} = 250$$

$$\text{Female students in college D} = 250 - 110 = 140$$

$$\text{Total students in college E} = 320 + 40 = 360$$

Number of students from college A, who have bank account in ICICI = 80% of $360 = 288$

Number of students from college B, who have bank account in ICICI = 75% of $340 = 255$

Number of students from college E, who have bank account in ICICI = 65% of $360 = 234$

Now, number of students from college A, who have bank account in PNB = $310 - 288 = 22$

Number of students from college B, who have bank account in PNB = $320 - 255 = 65$

Number of students from college E, who have bank account in PNB = $320 - 234 = 86$

In tabular form:

College	Number of students			Number of students who have banks accounts in		
	Total	Male	Female	ICICI	PNB	HDFC
A	360	225	135	288	22	50
B	340			255	65	20
C						30
D	390	250	140			60
E	360			234	86	40

81. (b) Number of male students in college B

$$= 225 \times \frac{8}{9} = 200$$

Number of male students in college E

$$= 225 \times \frac{5}{9} = 125$$

Then, number of female students in college B = $340 - 200 = 140$

Number of female students in college E

$$= 360 - 125 = 235$$

Then, difference between number of female students in college B and E

$$= 235 - 140 = 95$$

And number of students from college C and D together, who have bank accounts in HDFC

$$= 30 + 60 = 90$$

Therefore, required difference

$$= 95 - 90 = 5$$

82. (a) Number of students from college D, who have bank account in PNB

$$= 3 \times 77 - 65 - 86 = 80$$

Total number of students in college D

$$= 390$$

Now, number of students from college D, who have bank account in ICICI

$$= 390 - 80 - 60 = 250$$

Therefore, required percentage

$$= \left(\frac{250}{390} \right) \times 100 = 64\% \text{ (approx.)}$$

83. (d) Number of students in college B, who do not have account in ICICI

$$= 340 - 255 = 85$$

Number of students in college C, who do not have account in ICICI

$$= 226 - 85 = 141$$

Total number of students in college C

$$= 141 \times \frac{100}{100 - 70} = 470$$

Number of male students in college C

$$= \frac{470 + 60}{2} = 265$$

Number of female students in college A = 135

Therefore, required percentage

$$= \left(\frac{265 - 135}{135} \right) \times 100 = 96\% \text{ (approx.)}$$

Solution (84-87):

Table given below shows the created easy and difficult questions and total number of errors in easy and difficult questions:

Months	Created questions		Total number of errors	
	Easy questions	Difficult questions	Easy questions	Difficult questions
Jan	$60 \times \left(\frac{7}{12}\right) = 35$	$60 - 35 = 25$	$35 \times 0.6 = 21$	$25 \times 0.48 = 12$
Feb	$66 \times \left(\frac{8}{11}\right) = 48$	$66 - 48 = 18$	$48 \times 0.75 = 36$	$18 \times 0.5 = 9$
Mar	$70 \times \left(\frac{4}{7}\right) = 40$	$70 - 40 = 30$	$40 \times 0.5 = 20$	$30 \times 0.6 = 18$
Apr	$69 \times \left(\frac{15}{23}\right) = 45$	$69 - 45 = 24$	$45 \times 0.4 = 18$	$24 \times 0.25 = 6$
May	$66 \times \left(\frac{25}{33}\right) = 50$	$66 - 50 = 16$	$50 \times 0.64 = 32$	$16 \times 0.75 = 12$

84. (a) Total amount received by the freelancer in Jan = $(35 \times 40) + (25 \times 60) - (21 \times 3) - (12 \times 5) = 1400 + 1500 - 63 - 60 = ₹2777$
 Total amount received by the freelancer in Mar = $(40 \times 40) + (30 \times 60) - (20 \times 3) - (18 \times 5) = 1600 + 1800 - 60 - 90 = ₹3250$
 $\therefore$ Required sum = $2777 + 3250 = ₹6027$

85. (d) Total amount received by the freelancer in May for easy questions = $(50 \times 40) - (32 \times 3) = ₹1904$
 Total amount received by the freelancer in Feb for difficult questions = $(18 \times 60) - (9 \times 5) = ₹1035$
 $\therefore$ Required difference = $1904 - 1035 = ₹869$

86. (c) Total easy questions created in June = $(35 + 40 + 45 + 50) \times \frac{20}{100} = 34$
 Total number of errors = $34 \times \left(\frac{8}{17}\right) = 16$
 Earning in June = $34 \times 40 - 16 \times 3 = 1360 - 48 = ₹1312$

87. (b) Total increment in the earning = Penalty for errors in April = $18 \times 3 + 6 \times 5 = 54 + 30 = ₹84$
 Hence, required increment = ₹84

88. (d) Ratio time period = $\frac{P_1}{C_1} : \frac{P_2}{C_2}$
 Hence, ratio of their period of investment = $\left[\frac{x^3 + 8}{x^2 + 4 - 2x} \right] : \left[\frac{x^3 - 8}{x^2 + 4 + 2x} \right]$
 $= \left[\frac{(x+2)(x^2 + 4 - 2x)}{x^2 + 4 - 2x} \right] : \left[\frac{(x-2)(x^2 + 4 + 2x)}{x^2 + 4 + 2x} \right]$
 $= (x+2) : (x-2)$

89. (b) The pattern of the following series is:

91 $\underbrace{\quad 103 \quad 87 \quad 112 \quad 71 \quad 137 \quad 35}$
 $\underbrace{\quad +12 \quad -16 \quad +25 \quad -41 \quad +66 \quad -102}$
 $\underbrace{\quad +4 \quad +9 \quad +16 \quad +25 \quad +36}$

Hence, question mark should be replaced by 35.

90. (a) The pattern of the following series is:

$13 \times 5 + 4 = 69$
 $69 \times 5 - 3 = 342$
 $342 \times 5 + 4 = 1714$
 $1714 \times 5 - 3 = 8567$
 $8567 \times 5 + 4 = 42839$
 $42839 \times 5 - 3 = 214192$

Hence, question mark should be replaced by 342.

91. (a) Quantity I:

By the rule of alligation:

10% 20%
 $16\frac{2}{3}\%$

Ratio of labelled price of item A to that of

$B = \left(\frac{10}{3}\right) : \left(\frac{20}{3}\right) = 1 : 2$

Let labelled price of items A and B be 'a' and '2a' respectively.

Purchased price of items A and B together = ₹1000

$90\% \text{ of } a + 80\% \text{ of } 2a = 1000 \Rightarrow a = 400$

Labelled price of item A = a = ₹400

Labelled price of item B = 2a = ₹800

Labelled price of item C

= $1500 - 400 - 800 = ₹300$

Purchased price of item C

= $95\% \text{ of } 300 = ₹285$

Purchased price of all the three items together = $1000 + 285 = ₹1285$

Quantity II:

Let purchased price of items A, B, and C for Rahul be 72b, 128b, and 57b, respectively.

Labelled price of A = $72b \times \left(\frac{100}{90}\right) = 80b$

Labelled price of B = $128b \times \left(\frac{100}{80}\right) = 160b$

Labelled price of C = $57b \times \left(\frac{100}{95}\right) = 60b$

Total labelled price of all the three items together = $80b + 160b + 60b = 300b$

Now, $300b = 1500 \Rightarrow b = 5$

Labelled price of items A and B together = $80b + 160b = 240b = ₹1200$

Quantity III:

Let cost price of items A, B, and C be 32c, 50c, and 25c, respectively.

Labelled price of A = $125\% \text{ of } 32c = 40c$

Labelled price of B = $160\% \text{ of } 50c = 80c$

Labelled price of C = $120\% \text{ of } 25c = 30c$

Total labelled price of all the three items together = $40c + 80c + 30c = 1500$
 $\Rightarrow c = 10$

Cost price of all the three items together = $32c + 50c + 25c$

= $107c = ₹1070$

Quantity IV:

Let purchase price of item C = ₹d

Purchase price of B = ₹(d + 355)

Purchase price of A = ₹(d + 75)

Total labelled price of all the three items together = 1500

$\left[(d + 75) \times \left(\frac{100}{90}\right) \right] + [d + 355] \times \left(\frac{100}{80}\right) + \left[d \times \left(\frac{100}{95}\right) \right] = 1500$

$\Rightarrow d = 285$

Labelled price of A

= $(285 + 75) \times \left(\frac{100}{90}\right) = ₹400$

Labelled price of B

= $(285 + 355) \times \left(\frac{100}{80}\right) = ₹800$

Labelled price of C

= $285 \times \left(\frac{100}{95}\right) = ₹300$

Difference between labelled price of A and B = $800 - 400 = ₹400$

Difference between labelled price of B and C = $800 - 300 = ₹500$

Required sum = $1.4 \times (400 + 500) = ₹1260$

92. (e) Let time taken by P, Q, R, and S alone to finish the work be 'p', 'q', 'r', and 's' hours respectively.

Ratio of efficiency of P to Q = 2 : 3.

$\therefore p : q = 3 : 2$

$p = 1.5q$... (i)

$q - s = 5$

$\Rightarrow s = q - 5$... (ii)

According to the question:

$$\left(\frac{1}{p}\right) + \left(\frac{1}{q}\right) + \left(\frac{1}{r}\right) + \left(\frac{1}{s}\right) = \left(\frac{1}{6}\right) \quad \dots(iii)$$

$$\left(\frac{1}{r}\right) + \left(\frac{1}{s}\right) = \left(\frac{1}{12}\right) \quad \dots(iv)$$

From (iii) and (iv):

$$\left(\frac{1}{p}\right) + \left(\frac{1}{q}\right) = \left(\frac{1}{6}\right) - \left(\frac{1}{12}\right) = \frac{1}{12}$$

From equation (i):

$$\left(\frac{1}{1.5q}\right) + \left(\frac{1}{q}\right) = \frac{1}{12} \Rightarrow q = 20$$

$$p = 1.5q = 30$$

$$s = q - 5 = 15$$

From equation (iv):

$$\left(\frac{1}{r}\right) + \left(\frac{1}{15}\right) = \left(\frac{1}{12}\right) \Rightarrow r = 60 \text{ hours}$$

Quantity I:

Time taken by Q and R together to do the

$$\text{work} = \frac{1}{\left[\left(\frac{1}{20}\right) + \left(\frac{1}{60}\right)\right]} = 15 \text{ hours}$$

Quantity II:

Time taken by P alone to do the work when working with 80% of actual efficiency

$$= 30 \times \frac{5}{4} = 37.5 \text{ hours}$$

Time taken by R alone to do the work when working with 240% more than his actual efficiency

$$= 60 \times \left(\frac{100}{340}\right) = \left(\frac{300}{17}\right) \text{ hours}$$

Time taken by P and R together to do the

$$\text{work} = \frac{1}{\left[\left(\frac{1}{37.5}\right) + \left(\frac{17}{300}\right)\right]} = 12 \text{ hours}$$

Quantity III:

Let time taken by person T alone to do the work = 't' hours

Time taken by S and T together to do the

$$\text{work} = \frac{1}{\left[\left(\frac{1}{15}\right) + \left(\frac{1}{t}\right)\right]} = 8 \Rightarrow t = \frac{120}{7}$$

$$= 17.14 \text{ hours (approximately)}$$

Quantity IV:

Part of work done by P alone in 2 hours

$$= \frac{2}{30} = \frac{1}{15}$$

Part of work done by S alone in 1 hour

$$= \frac{1}{15}$$

Part of work done by P and S in 3 hours

$$= \left(\frac{1}{15}\right) + \left(\frac{1}{15}\right) = \left(\frac{2}{15}\right)$$

Part of work done by P and S in 21 hours

$$= \frac{14}{15}$$

Remaining work = $\frac{1}{15}$ which will be done by P alone in 2 hours

Time taken to finish the work by P and S = 21 + 2 = 23 hours

Quantity I > Quantity II < Quantity III < Quantity IV

93. (e) Age of Mukesh after 12 years

$$= 36 + 12 = 48 \text{ years}$$

Age of Manoj after 12 years

$$= 48 \times \left(\frac{5}{4}\right) = 60 \text{ years}$$

Present age of Manoj = 60 - 12 = 48 years

Present age of Manish = 48 - 18

= 30 years

Present age of Mayank = $30 \times \frac{7}{5} = 42$ years

Quantity I:

Average of present ages of all the four persons = $\frac{36 + 48 + 30 + 42}{4} = \frac{156}{4}$

= 39 years

Quantity II:

Oldest person is Manoj and youngest person is Manish.

Average present age of Manoj and Manish

$$= \left(\frac{48 + 30}{2}\right) = \frac{78}{2} = 39 \text{ years}$$

Quantity III:

Second oldest person is Mayank, and second youngest person is Mukesh.

Average present age of Mayank and

$$\text{Mukesh} = \left(\frac{42 + 36}{2}\right) = \frac{78}{2} = 39 \text{ years}$$

Quantity IV:

Sum of present ages of Mayank, Manish, Manoj, Mukesh, and Mahesh

$$= 36 \times 5 = 180 \text{ years}$$

Sum of present ages of Mayank, Manish, Manoj, and Mukesh

$$= 36 + 48 + 30 + 42 = 156 \text{ years}$$

Present age of Mahesh

$$= 180 - 156 = 24 \text{ years}$$

Quantity I = Quantity II = Quantity III > Quantity IV

94. (d) From I and II:

We know that, for two years

$$CI - SI = P \times \left(\frac{r}{100}\right)^2$$

$$\Rightarrow 256 = 40000 \times \left(\frac{r}{100}\right)^2$$

$$\Rightarrow 256 = 40000 \times \frac{r^2}{10000}$$

$$\Rightarrow r^2 = 64 \Rightarrow r = 8$$

We know that

$$CI = P \times \left(1 + \frac{r}{100}\right)^t - P$$

$$= 60000 \times \left(1 + \frac{8}{100}\right)^3 - 60000$$

$$= 60000 \times \frac{108}{100} \times \frac{108}{100} \times \frac{108}{100} - 60000$$

$$= 75582.72 - 60000 = ₹15582.72$$

I and II are not sufficient as we do not have any information about scheme B.

From I and III:

We know that,

$$CI = 60000 \times \left(1 + \frac{r}{100}\right)^3 - 60000$$

$$SI = \frac{P \times r \times t}{100} = \frac{45000 \times r \times 5}{100} = 2250 \times r$$

So, I and III are not sufficient.

From II and III:

$$SI = \frac{P \times r \times t}{100} = \frac{45000 \times 8 \times 5}{100}$$

[From statement 11, $r = 8$]

$$= ₹18000$$

II and III are not sufficient as we have no information about scheme A.

From I, II and III:

$$\text{Required difference} = 18000 - 15582.72$$

$$= ₹2417.28$$

Hence, all I, II and III together are sufficient.

95. (d) Let length and speed of train A are L m and S m/sec respectively.

From statement I:

Time taken by train A to cross a pole

$$= \left(\frac{L}{S}\right) \text{ sec}$$

Time taken by train A to cross a 60 m long

$$\text{bridge} = \left(\frac{L + 60}{S}\right)$$

According to question,

$$\frac{L + 60}{S} - \frac{L}{S} = 2 \Rightarrow \frac{60}{S} = 2 \Rightarrow S = 30$$

Hence, statement I alone is sufficient.

From statement II:

Length of train B = (L + 50) m

And speed of train B = 1.3S m/sec

Relative speed while moving in same direction = 1.3S - S = 0.3S m/sec

Relative speed while moving in opposite direction = 1.3S + S = 2.3S m/sec

According to question,

$$\left[\frac{2L + 50}{2.3S}\right] : \left[\frac{2L + 50}{0.3S}\right] = \frac{3}{23}$$

Not solvable, hence statement II alone is not sufficient.

From statement III:

Length of train C = (L + 100) m

According to question,

$$\frac{L + L + 100}{S} = \frac{L + 50}{(S - 15)} \Rightarrow 2S - 30 = S$$

$$\Rightarrow S = 30$$

Hence, statement III alone is sufficient.

Solutions (96-98):

Total players participated from country P in all the four games = $100 \times 4 = 400$
 Total players from country P participated in game $A = 20\%$ of $400 = 80$
 Total players from country P participated in game $B = 30\%$ of $400 = 120$
 Total players from country P participated in game $C = 10\%$ of $400 = 40$

Total players from country P participated in game $D = 40\%$ of $400 = 160$
 Total players participated from country Q in all the four games = $75 \times 4 = 300$
 Total players participated from country R in all the four games = $125 \times 4 = 500$
 Total players participated from country S in all the four games = $150 \times 4 = 600$
 Similarly, we can calculate other values as well:

Country	Total players	Players participated in game A	Players participated in game B	Players participated in game C	Players participated in game D
P	400	80	120	40	160
Q	300	25% of 300 = 75	15% of 300 = 45	40% of 300 = 120	20% of 300 = 60
R	500	25% of 500 = 125	30% of 500 = 150	20% of 500 = 100	25% of 500 = 125
S	600	25% of 600 = 150	40% of 600 = 240	5% of 600 = 30	30% of 600 = 180

96. (c) Number of medals will be minimum if number of players who won 2 medals each is minimum.
 Minimum number of players who won 2 medals each = $20 + 30 + 25 + 40 = 115$
 Total players who did not win any medal = 120
 Total players who won exactly 1 medal each = $400 - 115 - 120 = 165$
 Minimum number of medals won by country $P = 165 + 115 \times 2 = 395$

97. (a) Total medals won by country S in game $A = 30\%$ of $150 = 45$
 Total medals won by country S in game $B = m\%$ of $240 = 2.4m$
 Total medals won by country S in game $C = 20\%$ of $30 = 6$
 Total medals won by country S in game $D = (m + 20)\%$ of $180 = (1.8m + 36)$
 According to the question:
 $45 + 2.4m + 6 + (1.8m + 36) \leq 234$
 $\Rightarrow 4.2m \leq 147 \Rightarrow m \leq 35$

98. (a) Total players participated in game $C = 40 + 120 + 100 + 30 = 290$
 Female players participated in game C from country $P = x\%$ of $40 = 0.4x$
 Female players participated in game C from country $Q = (x + 10)\%$ of $120 = (1.2x + 12)$
 Female players participated in game C from country $R = y\%$ of $100 = y$
 Female players participated in game C from country $S = (y + 10)\%$ of $30 = (0.3y + 3)$
 Total male players = $290 - 0.4x - (1.2x + 12) - y - (0.3y + 3) = 162$
 $\Rightarrow 1.6x + 1.3y = 113$
 When $x = 30$: $y = 50$
 When $x = 20$: $y = 62.3$
 When $x = 10$: $y = 74.6$
 Hence, option (a) is true.

99. (d) Let the distance covered in AC car be x km.
 The distance covered in non AC car = $(200 - x)$ km
 According to the data provided in the question, we get:
 $x \times 14 + (200 - x) \times 6 = 1360$
 $\Rightarrow 14x + 1200 - 6x = 1360$
 $\Rightarrow 8x = 160 \Rightarrow x = 20$
 The distance covered in non- AC car = $(200 - 20) = 180$ km

Solutions (100-103):

Let the number of 1BHK, 2BHK and 3BHK flats available with dealer A be represented by p , q and r respectively.

For dealer A :

$q = 33.33\%$ of $(p + q + r)$

So, $(p + r) = (100 - 33.33)\%$ of the total

$$= \frac{2(p + q + r)}{3} \Rightarrow p + r = 680$$

$$\text{So, } (p + q + r) = 680 \times \frac{3}{2} = 1020$$

$$\text{So, } q = 1020 - 680 = 340$$

Similarly, we find the values for the other dealers:

Dealer	1BHK + 3BHK	2BHK	Total	Flats sold by dealer out of total flats sold by all dealers
A	680	340	1020	25%
B	790	210	1000	36%
C	650	350	1000	18%
D	390	390	780	21%
Total			3800	

100. (c) 50% of the 2BHK flats available with dealer D were sold.

Total number of flats sold by $D = 50\%$ of $390 + 90\%$ of $390 = 1.4 \times 390 = 546$

Now, D sold 21% of the total flats sold.

$$\text{So, total no. of sold flats} = \frac{546}{0.21}$$

$$\text{No. of flats sold by } A = \frac{546}{0.21} \times 0.25 = 650$$

$$\text{No. of 3BHK flats with } A = 680 - 320 = 360$$

$$\text{No. of 3BHK flats sold by } A = 35\% \text{ of } 360 = 126$$

$$\text{So, 1BHK and 2BHK flats sold by } A = 650 - 126 = 524$$

101. (d) No. of 2BHK flats sold by $A = 70\%$ of $340 = 238$

No. of 3BHK flats sold by A

$$= \left(\frac{7}{17}\right) \times 238 = 98$$

Also, total 3BHK flats available with

$$A = \frac{\text{3BHK flats sold}}{35\%} = \frac{98}{0.35} = 280$$

So, no. of 1BHK flats sold by A

$$= 41\% \text{ of } (680 - 280) = 164$$

Total no. of flats sold by A

$$= 238 + 98 + 164 = 500$$

Now, A sold 25% of the total flats sold.

Total number of flats sold by B

$$= \left(\frac{500}{0.25}\right) \times 0.36 = 720$$

102. (e) Total no. of flats sold by D

$$= 780 - 381 = 399$$

D sold 21% of the total flats sold.

So, total no. of flats sold by B

$$= \left(\frac{399}{0.21}\right) \times 0.36 = 684$$

No. of 2BHK flats sold by B

$$= 30\% \text{ of } 210 = 63$$

So, no. of 1BHK flats sold by B (as B did not sell any 3BHK flat) = $684 - 63 = 621$

Hence, no. of 1BHK flats available with B should be greater than 621 as no type of flat was sold 100%. So, only (I) is valid.

103. (a) No. of 3BHK flats with C

$$= \left(\frac{9}{13}\right) \times 650 = 450$$

No. of 3BHK flats sold by C

$$= \left(\frac{5}{7}\right) \times 350 = 250$$

C did not sell any 1BHK. As C has 350. 2BHKs, C can sell less than 350 2BHK flats. So, (I) is invalid.

If C sold 25, 2BHKs, then total number of flats sold by $C = 250 + 25 = 275$

As C sold 18% of total sold flats, so total

$$\text{no. of sold flats} = \frac{275}{0.18}$$

$$= 1527.77 \text{ (not a whole number)}$$

So, (II) is invalid as well.

If C sold 182, 2BHKs, then total number of flats sold by $C = 250 + 182 = 432$

$$\text{So total no. of sold flats} = \frac{432}{0.18} = 2400$$

So, only (III) is valid.

Solutions (104-106):

There are 80 teachers in school *A* and *C* together which are 50 more than the number of male teachers in school *B*.

∴ Number of male teachers in school *B*

$$= 80 - 50 = 30$$

40% of teachers in school *B* are females.

Then, Total teachers in school *B*

$$= B = 30 \times \frac{100}{100 - 40} = 50$$

Number of female teachers in school *B*

$$= 50 - 30 = 20$$

Number of students in school *B* = $50 \times 16 = 800$

65% of the students in school *B* are females. Then,

Number of female students in school *B*

$$= 65\% \text{ of } 800 = 520$$

Number of male students in school *B*

$$= 800 - 520 = 280$$

Number of students in school *C*

$$= 700 + 800 = 1500$$

Number of students in school *A*

$$= (100 - 16)\% \text{ of } 1500 = 1260$$

Number of male teachers in school *C*

$$= 3\% \text{ of } 800 = 24$$

Number of female teachers in school *C*

$$= 24 - 4 = 20$$

Total teachers in school *C* = $24 + 20 = 44$

Number of male teachers in school *A* to *C* is in the ratio 1 : 3 respectively. Then,

Number of male teachers in school *A*

$$= 24 \times \frac{1}{3} = 8$$

Number of female teachers in school *A*

$$= 8 \times \frac{7}{2} = 28$$

Total teachers in school *A* = $8 + 28 = 36$

Only one member was present at the fest from family of students in school *A*.

∴ Total family members present in school *A*

$$= 1 \times 1260 = 1260$$

Total family members present in school *C*

$$= 4260 - 1260 = 3000$$

Total family members present in school *B* = $1 \times 35\% \text{ of } 800 + 2 \times (100 - 35)\% \text{ of } 800 = 1320$

In tabular form:

School	Number of teachers			Number of students			Number of family members present at the fest
	Total	Male	Female	Total	Male	Female	
A	36	8	28	1260			1260
B	50	30	20	800	280	520	1320
C	44	24	20	1500			300

104. (b) Number of female students in school *A*

$$= 28 \times \frac{25}{2} = 350$$

Number of male students in school *A*

$$= 1260 - 350 = 910$$

Number of male students in school *D*

$$= 910 + 50 = 960$$

Total students in school *D*

$$D = 960 \times \frac{100}{100 - 40} = 1600$$

Total students in school *C* = 1500

Therefore, required difference

$$= 1600 - 1500 = 100$$

105. (a) We know,

$$\text{average} = \frac{\text{sum of observations}}{\text{number of observations}}$$

Let the value for the blank = n

Sum of age of male teachers in school *A*

$$= 8 \times 24 = 192$$

Sum of age of male teachers in school *B*

$$= 30n$$

Sum of age of male teachers in school *C*

$$= 24 \times 20 = 480$$

Sum of age of male teachers in schools *A*, *B* and *C* = $(8 + 30n + 24) \times 21 = 1302$

Then, $1302 = 192 + 30n + 480$

$$\Rightarrow 30n = 630 \Rightarrow n = 21$$

106. (d) In school *C*:

Family members present at the fest from 80% of students' family

$$= 1 \times 80\% \text{ of } 1500 = 1200$$

And family members present at the fest from rest 20% of students' family

$$= (k - 2) \times 20\% \text{ of } 1500 = 300k - 600$$

$$\text{Now, } 3000 = 1200 + 300k - 600$$

$$\Rightarrow 300k = 2400 \Rightarrow k = 8$$

Solutions (107-109):

Upstream speed of boat *L* = $12 - 4 = 8$ km/h

And downstream speed of the boat *L*

$$= 12 + 4 = 16 \text{ km/h}$$

Speed of boat *K* in still water = $12 + 4 = 16$ km/h

Downstream speed of the boat *K*

$$= 16 + 4 = 20 \text{ km/h}$$

$$\text{Upstream speed of boat } M = \frac{35}{2.5} = 14 \text{ km/h}$$

And speed of boat *M* in still water

$$= 14 + 4 = 18 \text{ km/h}$$

Thus, downstream speed of boat *M* = $18 + 4$

$$= 22 \text{ km/h}$$

Distance travelled by boat *L* in upstream

$$= 35 - 3 = 32 \text{ km}$$

So, time taken by boat *L* in upstream

$$= \frac{32}{8} = 4 \text{ hours}$$

Time taken by boat *M* to travel in downstream = $4.5 - 2.5 = 2$ hours

So, distance travelled in downstream by boat *M*

$$= 2 \times 22 = 44 \text{ km}$$

Time taken in downstream by boat *L*

$$= \frac{48}{16} = 3 \text{ hours}$$

The distance travelled by boat *K* in downstream = $48 - 18 = 30 \text{ km}$

Thus, time taken in downstream by boat *K*

$$= \frac{30}{20} = 1.5 \text{ hours}$$

And time taken in upstream by boat *K*

$$= 5 - 1.5 = 3.5 \text{ hours}$$

Hence, distance covered by boat *K* in upstream = $3.5 \times 12 = 42 \text{ km}$

107. (e) Required ratio = $42 : 44 = 21 : 22$

$$108. (d) \text{ Required \%} = \frac{4 - 3.5}{4} \times 100$$

$$= 12.5\% \text{ less}$$

109. (d) Required difference = $32 - 30$

$$= 2 \text{ km less}$$

Solutions (110-113):

The number of candidates, who got interview call for 3 posts in company *A* = $30 \times 3 = 90$

The number of candidates, who got interview call for 6 posts in company *B* = $3x \times 6 = 18x$

The number of candidates, who got interview call for 1 post in company *C* = $2x$

According to the data provided in the question, we get

$$18x - 2x = 160 \Rightarrow 16x = 160 \Rightarrow x = 10$$

The number of candidates, who got interview call for 6 posts in company *B*

$$= 18 \times 10 = 180$$

The number of candidates, who got interview call for 1 post in company *C*

$$= 2 \times 10 = 20$$

Let the total number of candidates applied in company *B* = $4a$

∴ The total number of candidates applied in company *C* = a

The total number of candidates applied in company *A* = $\frac{5a}{3}$

110. (d) The number of candidates who applied in company *B* can't be determined.

111. (c) Required percentage

$$= \frac{a}{5a} \times 100 = 60\%$$

112. (c) The total number of candidates who got interview call from company *A* = 90

The number of female candidates who got interview call from company *A*

$$= 90 \times 0.70 = 63$$

The number of female who got job in

$$\text{company } A = \frac{1}{3} \times 3 = 1$$

The number of females who did not get the job in company *A* = $63 - 1 = 62$

113. (d) According to the data provided in the question, we get:

$$\frac{5a}{3} - a = 150$$

$$\Rightarrow \frac{5a - 3a}{3} = 150 \Rightarrow 2a = 450 \Rightarrow a = 225$$

The total number of candidates who applied in company $B = 4 \times 225 = 900$

The number of candidates who did not get interview call from company $B = 900 - 180 = 720$

114. (a) The probability of success of A

$$A = \frac{1}{1+4} = \frac{1}{5}$$

The probability of failure of A

$$A = \frac{4}{1+4} = \frac{4}{5}$$

The probability of failure of A

$$B = \left(\frac{4}{5}\right) \times \frac{375}{4 \times 100} = \frac{3}{4}$$

The probability of success of B

$$B = 1 - \frac{3}{4} = \frac{1}{4}$$

The probability of success of C

$$C = \left(\frac{1}{5}\right) \times \frac{500}{3 \times 100} = \frac{1}{3}$$

The probability of failure of $C = 1 - \frac{1}{3} = \frac{2}{3}$

The probability that no one will succeed in the examination-

$$= \left(\frac{4}{5}\right) \times \left(\frac{3}{4}\right) \times \left(\frac{2}{3}\right) = \frac{2}{5}$$

The probability that no one will fail in the examination-

$$= \left(\frac{1}{5}\right) \times \left(\frac{1}{4}\right) \times \left(\frac{1}{3}\right) = \frac{1}{60}$$

Required ratio $= \frac{2}{5} : \frac{1}{60} = 24 : 1$

115. (c) According to question,

$$30\% \times 40 + Y\% \times Z = 50\% \times (Z + 40)$$

$$12 + \frac{YZ}{100} = 50\%Z + 20$$

$$YZ - 50Z = 800 \quad \dots(i)$$

Also,

$$90\% \times Z + 30\% \times 40 = (Y + 24)$$

$$90\% \times Z = Y + 24 - 12$$

$$Z = 10/9 \times (Y + 12) \quad \dots(ii)$$

From (i) and (ii)

$$\frac{10}{9} \times (Y + 12) \times (Y - 50) = 800$$

$$Y^2 - 38Y - 1320 = 0$$

$$(Y - 60)(Y + 22) = 0$$

So, possible value of $Y = 60$

$$\text{Value of } Z = \frac{10}{9} \times (60 + 12) = 80$$

Hence, total number of questions in the exam $= 40 + 80 = 120$

116. (b) In September 2022, the Monetary Policy Committee of Reserve Bank of India decided to increase the Bank Rate to 6.15%.

The Reserve Bank of India (RBI) doesn't directly set a "Bank Rate." It uses the Repo Rate to control liquidity.

The Reserve Bank of India (RBI) has raised the repo rate by 1.9 percent since May 2022.

117. (a) Minicoy, Thundi Beach, and Kadmat Beach are located in the Union Territory of Lakshadweep.

Lakshadweep is a group of islands in the Arabian Sea, off the southwestern coast of India.

The islands are known for their pristine beaches, turquoise waters, and rich marine biodiversity.

The Blue Beaches eco-label is awarded to the cleanest beaches in the world, and these two beaches in Lakshadweep have earned this prestigious recognition for their cleanliness and environmental sustainability efforts.

118. (c) The web-based application launched by RBI Governor Shaktikanta Das on October 6, 2022, is named "DAKSH."

DAKSH is designed to enhance supervisory processes, aiming to improve the efficiency and effectiveness of regulatory oversight.

This application likely integrates advanced technological features to streamline communication, data analysis, and risk assessment within the supervisory framework.

The introduction of DAKSH reflects the Reserve Bank of India's commitment to modernizing its supervisory practices and ensuring the stability and resilience of the financial system.

119. (e) Spanish tennis player Carlos Alcaraz won the 2022 US Open. It was the first Grand Slam title of his career, achieved before his Wimbledon victory in 2023.

The 20-year old is the ninth male singles winner from Spain and the first new champion since Rafael Nadal in 2005.

In the men's singles final, the Spanish player defeated Casper Ruud of Norway 6-4, 2-6, 7-6, 6-3.

Alcaraz became the youngest player since Pete Sampras in 1990 to win the US Open men's singles title."

120. (d) Rishi Sunak became the 57th Prime Minister of the United Kingdom on October 25, 2022.

He succeeded Liz Truss, becoming the third Prime Minister of the UK in that year.

He is also the first person of color and the first practicing Hindu to hold this position. At 42, he was the youngest Prime Minister in two centuries. His appointment marked a significant moment in UK political history.

121. (a) On 19 November 2022, Prime Minister Narendra Modi inaugurated the Donyi Polo Airport at Itanagar in Arunachal Pradesh.

It is the first Greenfield Airport in the state.

It is equipped with ultra-modern facilities to ensure all-weather operations.

The name of the airport is given after Donyi and Polo, i.e. Sun and Moon.

122. (c) As per the Human Development Report 2021-22, India's position on the Human Development Index (HDI) declined from 130th in 2020 to 132nd in 2022.

Switzerland has been ranked top among 191 countries in HDI 2022.

European countries dominated the rankings, with 8 out of the top 10 positions held by nations from the region.

Sri Lanka emerged as the best performer in the Indian sub-continent.

Since 1990, Human Development Reports (HDRs) have been published, focusing on various themes through the lens of the human development approach.

123. (a) The leaders of the Shanghai Cooperation Organisation (SCO) endorsed the holy city of Varanasi as the first ever tourist and cultural capital of the grouping for the year 2022-23.

The Indian delegation at the SCO summit in this historic Uzbek city of Samarkand was led by Prime Minister Narendra Modi.

Uzbekistan handed over the rotating presidency of the grouping to India for the year 2022-23.

124. (b) On October 30, 2022, Prime Minister Shri Narendra Modi dedicated the nation's first private-sector transport aircraft manufacturing plant, the C-295, in Vadodara, Gujarat.

Tata Advanced Systems Limited and Airbus Defence and Space S.A., Spain are collaborating to manufacture C-295 aircraft for the Indian Air Force (IAF) at this site.

This is the first initiative of its sort where a commercial enterprise will produce a military aircraft in India.

The project would cost a total of Rs 21,935 crore. There are many civilian uses for the aircraft.

125. (b) The US economists Douglas Diamond, Philip H. Dybvig, and Ben S. Bernanke have been awarded the 2022 Nobel Prize in Economics.

The research conducted by the trio lessens the possibility that financial crises would turn into protracted depressions with dire ramifications for society.

The work of Bernanke Ben Bernanke, the former head of the US Federal Reserve, examined the 1930s Great Depression.

He demonstrated, among other things, how bank runs played a critical role in the crisis's extreme depth and protracted duration.

126. (a) The 14th World Spice Congress (WSC) was organized by the Spices Board India in collaboration with various trade and export forums.

- It took place in Navi Mumbai, Maharashtra, from February 16 to 18, 2023.
- The theme of the event was “Vision 2030: Spice Industry Ready for the Future.”
- The Congress brought together spice producers, traders, policymakers, and researchers from across the globe.
127. (b) In November 2022, the Reserve Bank of India (RBI) imposed a monetary penalty of ₹1,76,00,000 (Rupees One Crore and Seventy Six Lakh only) on Vakrangee Limited for non-compliance with certain provisions of the White Label ATM (WLA) directions.
- The penalty was in respect to non-compliance with regulations on ATM deployment targets.
- This action was based on deficiencies in regulatory compliance.
128. (a) In order to settle trade in Indian rupees, UCO Bank has been given permission by the RBI to open a special vostro account with Gazprombank of Russia.
- As a result, it is now the first bank to be approved by the RBI to settle trades in Indian currency.
- RBI published a circular on July 11, 2022, permitting commercial settlements in rupees between India and other nations.
129. (d) The time required for the listing of such developing investment vehicles was cut by the market regulator SEBI from the current 12 days to six working days.
- The choice was made to expedite the public offering of REIT and InvIT units.
- A public offering of Real Estate Investment Trust (REIT) and Infrastructure Investment Trust (InvIT) units that opens on June 1st or later will be subject to the new regulation.
- The goal of the action is to streamline and reduce the cost of the current process for issuing these types of developing investment vehicles.
130. (d) In the Puri region of Odisha, at Konark's Chandrabhaga Beach, the ninth edition of the International Sand Art Festival has commenced.
- The festival's chief curator is the well-known sand artist and Padmashree Awardee, Sudarsan Pattnaik.
- The first week of December is when the celebration is annually commemorated with considerable enthusiasm.
- Amazing sand artwork created by talented artists from India and other countries is on show at the sand art festival.
131. (c) The ICICI Prudential Nifty Auto Index Fund is the first auto index fund in India, introduced by ICICI Prudential Mutual Fund.
- It is an index scheme that is open-ended and replicates the Nifty Auto Index.
- It is intended to mirror the actions and outcomes of the financial market's car sector.

132. (e) In collaboration with National E-Governance Services Limited (NeSL), HDFC Bank, the biggest private sector bank in India, issued the nation's first Electronic Bank Guarantee (e-BG).
- With the new electronic bank guarantees, which can be processed, stamped, validated, and given promptly with greater security, the laborious, paper-based process has been abolished.
- e-BG was created with input from the CVC-CBI committee, NeSL, and IBA in order to streamline the procedure and get rid of any chance for fraud or manipulation.
133. (d) The Coca-Cola Company produced the transparent, lime- and lemon-flavored soft drink known as Sprite, which is owned by Americans.
- The Coca-Cola Company purchased the rights to the Sprite brand name in 1960.
- T. C. Bud Evans, a Houston-based bottler who also distributed Coca-Cola products, came up with the name about 1955 for a line of drinks that included flavors like orange and strawberry.
- Originally known as Fanta Klare Zitronen (or “Fanta Clear Lemon” in English), the lemon-lime beverage that is now known as Sprite was created in West Germany in 1959 and debuted as a 7 Up rival in the United States in 1961 under the Sprite brand.
134. (a) With the designation of “Maharatna” Central Public Sector Enterprise, Rural Electrification Corporation Limited (REC) now enjoys increased operational and financial independence.
- On September 21, the Department of Public Enterprises, a division of the Ministry of Finance, issued an order to this effect.
- REC is an NBFC that was founded in 1969 and is dedicated to the financing and development of the power sector in India.
135. (b) The Tamil Nadu Legislative Assembly passed the Tamil Nadu Prohibition of Online Gambling and Regulation of Online Games Act, 2022 to ban online gambling and regulate online games.
- The bill was introduced to address the growing concerns over online gaming addiction and its impact on public health and safety.
- It aims to prohibit games involving wagering or betting, including online rummy and poker.
136. (d) The Competition Commission of India (CCI) approved the merger between Sony Pictures Networks India (SPNI), now

- known as Culver Max Entertainment Pvt. Ltd., and Zee Entertainment Enterprises Ltd. (ZEEL) in October 2022.
- The merger aimed to create one of India's largest media conglomerates, combining their extensive television networks, digital platforms, and content libraries.
- Zee, a household TV name in India, was set up in 1992 by Subhash Chandra, dubbed the ‘Father of Indian Television’.
137. (d) Social Stocks Exchange (SSE) is a new idea in India; the goal of the course is to channel more funds towards the non-profit and private sectors.
- In her budget statement for the 2019–20 fiscal year, Finance Minister Nirmala Sitharaman initially proposed the concept of SSE.
- As required by SEBI regulations, the Listed NPO must provide SSE with a statement on the use of funds within 45 days of the end of the quarter.
- Social enterprises that use SSE to raise money have been asked by SEBI to release their Annual Impact Report (AIR) within ninety days of the end of the fiscal year.
138. (c) On September 17, the National Logistics Policy was introduced in New Delhi by Prime Minister Shri Narendra Modi. The National Logistics Policy was another significant step that India had taken toward becoming a developed nation.
- The strategy aims to bring India's logistics costs down to levels that are competitive with international benchmarks by 2030, rank in the top 25 of the Logistics Performance Index by then, and establish a data-driven decision support system for an effective logistics ecosystem.
- The National Logistics Policy's adoption would lower logistics costs, promote foreign trade, assist in making India “atmanirbhar,” or self-reliant, usher in prosperity for the country, and open up new chances for our startups.
139. (d) As per Reserve Bank of India (RBI) guidelines, Commercial Papers (CPs) are issued in a minimum denomination of Rs. 5 lakh and in multiples of Rs. 5 lakh thereafter.
- The tenor of CPs ranges from 7 days to 1 year, and they are issued at a discount to face value.
- Commercial Papers (CPs) are unsecured, short-term debt instruments issued by companies to raise funds for a short duration, usually up to one year.
- It was initially established in India in 1990 and is an unsecured money market instrument issued as a promissory note.

140. (a) In October 2022, the GEAC, under the Ministry of Environment, Forest and Climate Change, approved the environmental release of Dhara GM mustard (DMH-11) for seed production and further testing.
DMH-11 was developed by the Centre for Genetic Manipulation of Crop Plants, University of Delhi, using the barnase-barstar gene system to enable male sterility and hybrid seed production.
This marks the first ever GM food crop for commercial release in India, following Bt cotton.
141. (c) On 11 November 2020, the Cabinet approved the launch of the Production Linked Incentive (PLI) Scheme for 10 key sectors to boost India's manufacturing capabilities and promote exports.
It aims to promote manufacturing of high efficiency solar PV modules in India and thus reduce import dependence in the area of Renewable Energy.
The financial outlay for PLI for 'High Efficiency Solar PV Modules' (Tranche-I) is ₹ 4,500 crore.
There are now 14 productivity-linked incentive (PLI) sectors in the Union Budget 2022–2023.
142. (d) Among the options provided, the entity that typically cannot open a savings account in a Scheduled Commercial Bank is the State Government.
The State Government may not be able to open a savings account directly in the name of the government itself, as banking regulations and practices often require specific procedures for government entities to handle their finances, which may involve separate government bank accounts or treasury accounts managed by designated agencies.
143. (b) Bimal Julka succeeded Sudhir Bhargava as Chief Information Commissioner in March 2020.
He was selected by the high-powered selection committee headed by Narendra Modi.
144. (c) For Rs 55 crore, Punjab & Sind Bank has agreed to purchase a 2% share in the asset rebuilding company NARCL.
The equity investment would be made in installments, with the completion of the first tranche, estimated to cost Rs 28.18 crore, by March 31, 2022.
The money is being invested in order to guarantee quick resolution, which includes recovering non-performing assets (NPA) accounts.
145. (c) The Union Budget 2022–2023 was presented by Smt. Nirmala Sitharaman, Union Minister of Finance and Corporate Affairs.
Out of all the major economies, India's economic growth is expected to be the greatest at 9.2%. Therefore, statement 1 is true.
- The productivity linked incentive scheme is expected to generate 60 lakh new jobs across 14 sectors.
The portals for ASEEM, NCS, e-shram, and Udyam will be connected. Therefore, statement 1 is true.
146. (a) The Reserve Bank of India, the nation's central bank, issues digital currency notes known as Digital Rupee (₹), or Central Bank Digital Currency (CBDC).
The first official announcement on the much-awaited launch of the digital currency came from Union Finance Minister Nirmala Sitharaman, who stated before that the RBI will introduce a CBDC in 2022–2023.
CBDC: Digital tokens produced by central banks that resemble cryptocurrencies are known as central bank digital currencies.
CBDCs are being developed in many nations, and some have already put them into practice.
147. (a) The Singaporean sovereign investor GIC spearheaded a \$51 million fundraising round for the Indian rocket firm Skyroot Aerospace.
By the middle of next year, the firm intends to use it for commercial satellite launches.
The funding comes as the Indian government pushed for the private sector to support the nation's state-run space program, which is well-known for its reasonably priced launches and missions.
The first privately made space launchers in India are being produced by Skyroot; they have an all-carbon fiber frame and can carry payloads up to 800 kg (1,760 pounds) into low-Earth orbit.
It already has a facility and expertise exchange arrangement with the government-run Indian Space Research Organization (ISRO).
148. (d) FIBAC 2022, the annual banking conference, was jointly organized by the Federation of Indian Chambers of Commerce and Industry (FICCI) and the Indian Banks' Association (IBA).
FIBAC (FICCI-IBA Conference) is a premier platform for dialogue between financial institutions, policymakers, and business leaders.
The conference featured participation from senior leaders of public and private sector banks, NBFCs, fintech companies, and regulators.
149. (b) The "Mission Schools of Excellence" project was introduced by Prime Minister Modi in Gandhinagar, Gujarat, on October 19, 2022.
The World Bank provides a portion of the Rs 10,000 crore total budget that went into the mission's conception.
By installing new, smart classrooms, computer laboratories, and general school infrastructure upgrades, it will support the development of Gujarat's educational system.
150. (d) For the first time, SVC Co-operative Bank Ltd. and Small Industries Development Bank in India (SIDBI) have teamed up to support MSMEs.
In accordance with the agreement, SIDBI will provide SVC Bank with access to the refinance facility in order to enhance the flow of loans to MSMEs.
This is the first time a UCB has signed such an arrangement. More of these agreements will be signed by SIDBI with RRBs and UCBs in different states.
151. (a) The Indian government issued sovereign green bonds worth ₹16,000 crore (approximately \$3.3 billion) during the entire FY23, not just the first half.
While there isn't official data on the exact split between halves, it's safe to assume it wasn't issued equally throughout the year.
It's important to note that India plans to raise ₹20,000 crore (\$4 billion) through sovereign green bonds in FY 2023-24.
152. (b) Large modern currency chests are now able to charge higher service fees—up to ₹8 per packet—for cash deposited by non-chest bank branches. Previously, these fees were set at ₹5 each packet of 100 pieces. This increase was permitted by the Reserve Bank of India.
Because of this, a currency chest (CC) will only be considered a large modern CC if it meets the required minimum specifications.
153. (a) For creating the greatest Enterprise Payments Hub (EPH) in the world, IndusInd Bank has won the global "Celent Model Bank" award.
It has emerged victorious in the 'Payments System Transformation' division.
The Bank was recognized with this award for its exceptional efforts in building a cloud-based central payments hub that can handle large volumes of transactions coming from all types of payment instructions and all first client touchpoints.
The "Celent Model Bank" is given out for exemplary technology use in a variety of domains that are essential to banking performance.
154. (b) Indian Overseas Bank (IOB) was hit with a ₹57.50 lakh fine by the Reserve Bank of India (RBI) on June 24, 2022, for failing to follow certain guidelines.
The fine was in relation to the RBI's guidelines on frauds: Classification and Reporting by Certain Financial Institutions and Commercial Banks.
155. (b) The Laxmi Co-operative Bank Ltd.'s license was revoked by the Reserve Bank of India (RBI) on September 22, 2022, for insufficient capital and earning potential.
Depositors will thereafter be eligible to collect the full amount of their deposit insurance claim, up to a maximum of Rs 5 lakh.
The RBI also revoked Rupee Co-operative Bank Ltd.'s license, which is located in Pune, on August 10, 2022.